

Introduction of Geographical Information System

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Chapter 2: Spatial analysis

- 1. Spatial analysis concept**
- 2. Topology rules**
- 3. Buffers, Overlays, and Spatial Joins**
- 4. Raster analysis**

1. Spatial analysis concept

(Material based on Tadele Feyssa, Wollega University)

- Analysis is the process of inferring meaning from data.
- Analysis is carried visually in a GIS
- Analysis in a GIS can also be carried out by measurements, statistical computations, fitting models to data values and other operations

Spatial analysis concept

- Spatial analysis: the process by which we turn raw data into **useful information**
- Spatial analysis: the transformations, manipulations, and methods applied to geographic data to add values, to support decisions, and to reveal patterns and anomalies that are not immediately obvious.
- In a narrow sense: a method for **analyzing spatial data**, while in a broad sense it includes **revealing and clarifying processes, structures**, etc., of spatial phenomena that occur on the Earth's surface.
- Spatial analysis: designed to support **spatial decision making**, and to serve as a tool for assisting with regional planning and the formulation of government policies, among other things.

Spatial analysis



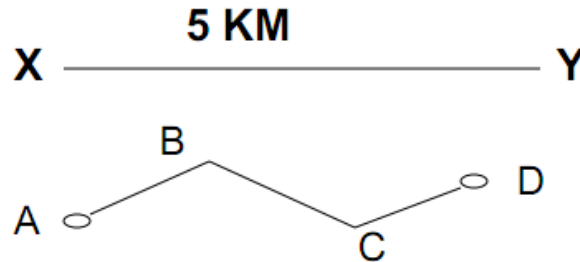
Spatial Analysis Functions

- *Measurements*
- *Query*
- *Extraction*
- *Proximity*
- *Classification*
- *Topology*
- *Network analysis*

Measurement

Measurements are simple numerical values that describe aspects of geographic data.

DISTANCE



$$A - B = 20 = 40\%$$

$$B - C = 20 = 40\%$$

$$C - D = 10 = 20\%$$

Perimeter



AREA/SIZE



Queries

- *Queries*: basic operations to answer simple questions posed by the user.
- **No changes** occur in the database, and **no new data** are produced
- Picking features based on query expressions, which **use Boolean algebra (and, or, not), set algebra** ($>$, $<$, $=$, $>=$, $<=$), arithmetic operators ($=$, $-$, $*$, $/$), and user-defined values.
- Ex. 'How many houses are found within 1 km of this point', or 'which is the closest city to Ha Noi going north'.

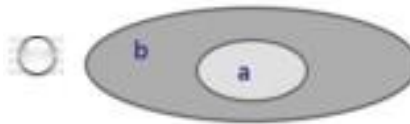
Intersects



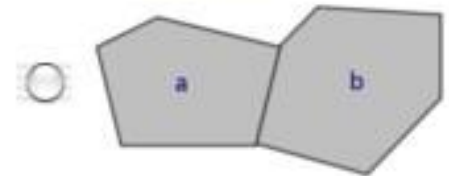
Distance (buffer)



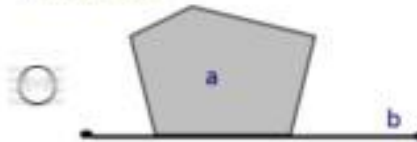
Within(a,b)



Touches(a,b)



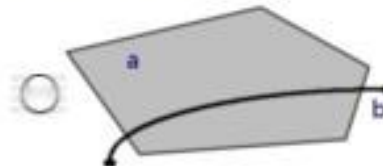
Touches(a,b)



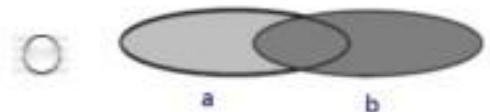
Crosses(a,b)



Crosses(a,b)



Overlaps(a,b)



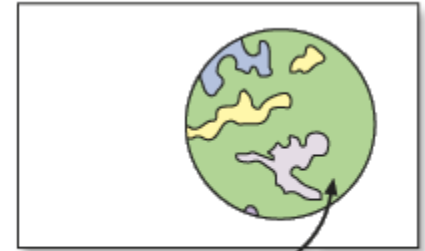
Extraction

- Extracting portions of data to **isolate specific areas** for further processing or data analysis.
- Extraction functions **reduce the size** of datasets and/or facilitate more complex interpretation.
- GIS tools to extract data: clip, select, and split
- Extraction by location,
Extraction by shapes,
Extraction by attribute

Extraction by shapes



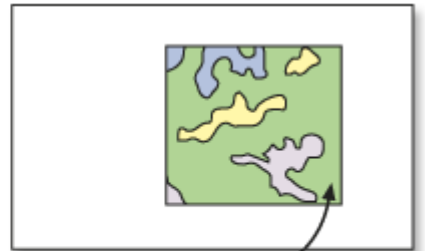
Specified circle



Selected cells for processing within the analysis window

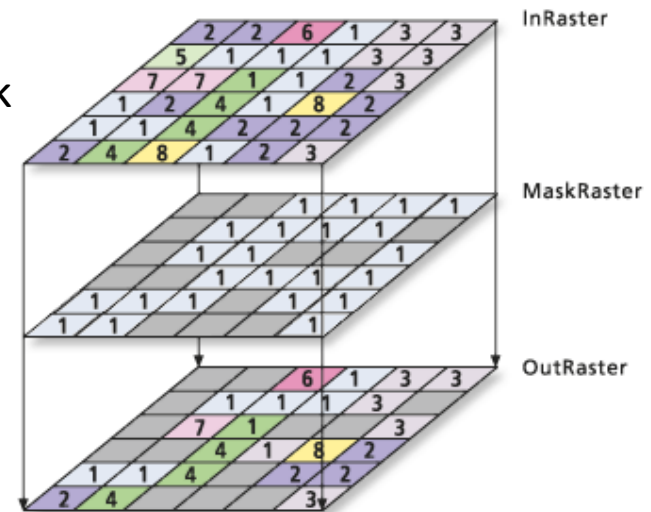


Specified rectangle



Selected cells for processing within the analysis window

Extraction by mask



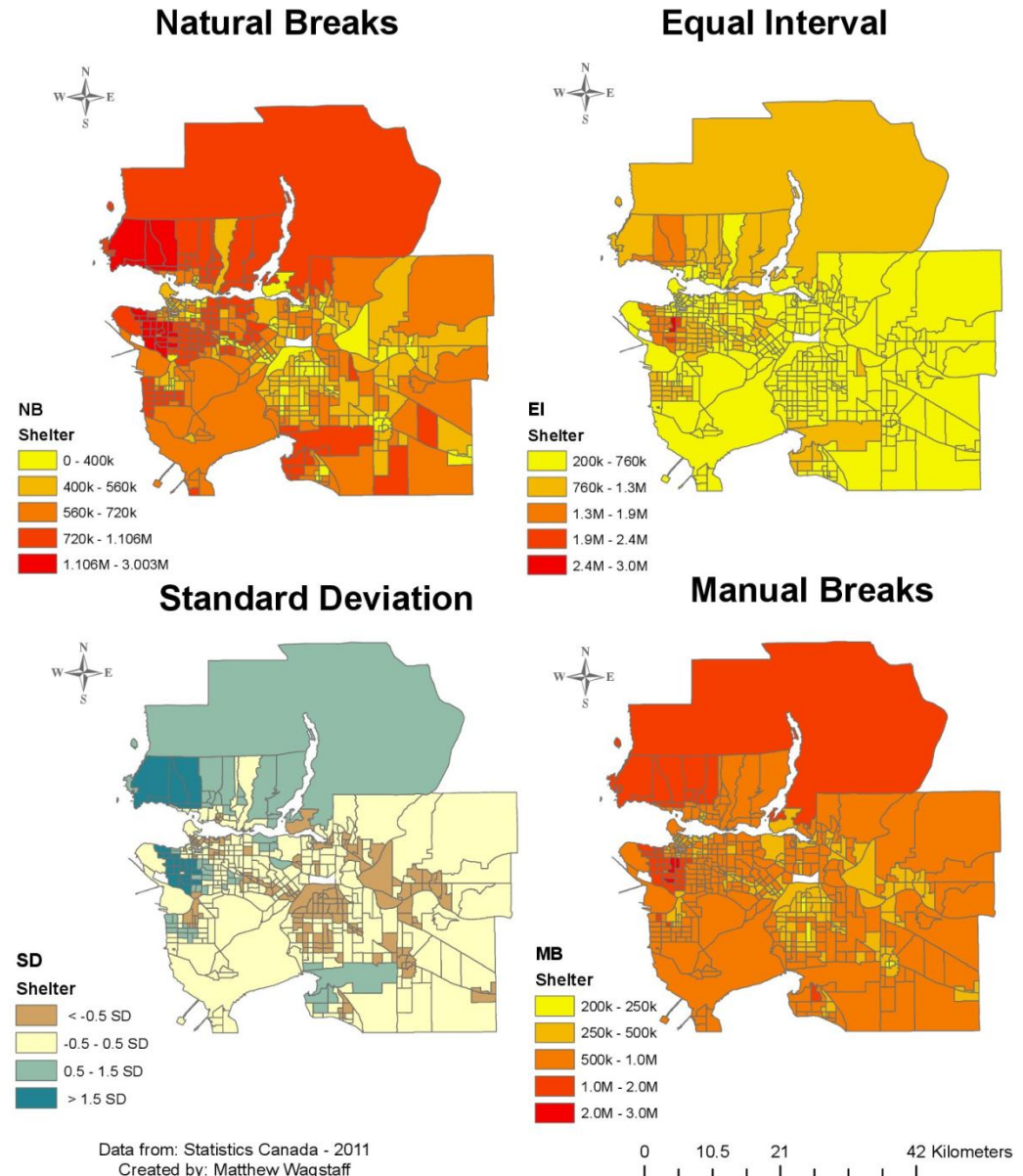
Classification

- The procedure of identifying a **set of features** as belonging to a group and defining patterns.
- Can defines **patterns**.
- Assisting in recognizing **new patterns**.
- Using **single data layers**, or with **multiple data** layers
- Aggregating features into classes allows **to spot patterns** in the data more easily.
- By altering the **class breaks** (the boundary between classes), we can create very different-looking maps.
- Classes can be created manually, or you can use a standard classification scheme.

Classification (Methods)

- **Natural Breaks**
 - Seeks to minimize the average deviation from the class mean while maximizing the deviation from the means of the other groups
- **Equal Interval**
 - Based on **the range** of the data and the number of bins you choose.
- **Standard Deviations**
 - Based on standard deviations of the values
- **Manual Breaks**
 - Select the breaks to bias the presentation of the data to meet your purpose.

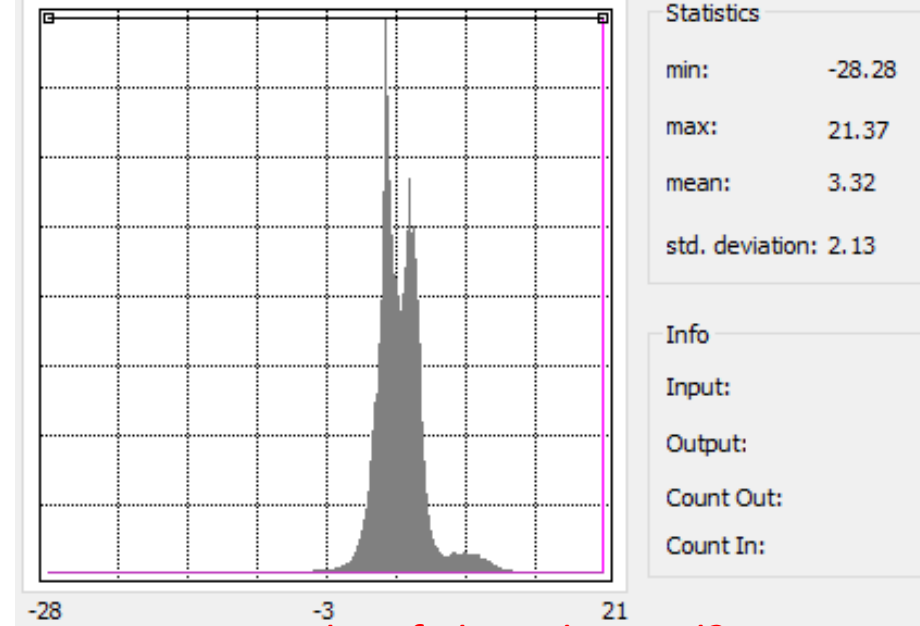
Comparing Classification Methods of Shelter Cost in Vancouver Census Tracts



Classification

(Statistical numbers)

- Histogram



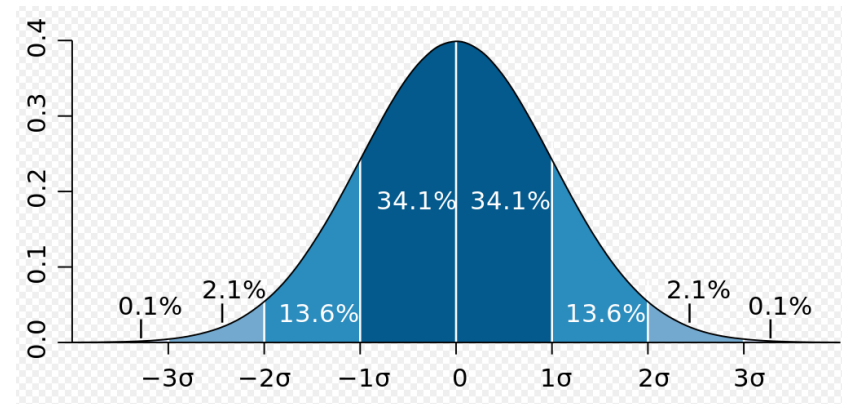
Practice to classify based on Std?

- Mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

- Standard deviation

$$S = \sqrt{S_{n-1}^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$



Normal distribution (or bell-shaped curve) where each band has a width of 1 standard deviation – See also: 68–95–99.7 rule ([Petter Strandmark](#))

Topology

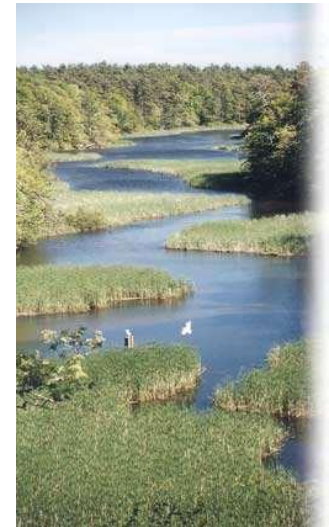
- **Topology** is the arrangement that **defines** how point, line, and polygon features share coincident geometry.
- For example, street centerlines and census blocks share common geometry, and adjacent soil polygons share their common boundaries

Proximity computation

- Using geometric distance to define the **neighborhood** of one or more target locations
- Proximity includes: buffering, interpolation, Thiessen polygons, and various topographic functions.

Network Analysis

- Network is *any system of interconnected linear features*
- A network is a system of **interconnected elements**, such as edges (lines) and connecting junctions (points), that represent possible routes from one location to another
- Solving problems involving networks
- Its goal is efficiency – Saving time and money.



Network Analysis

- To solve common network problems:
 - finding the best route across a city,
 - finding the closest emergency vehicle or facility
 - identifying a service area around a location
 - servicing a set of orders with a fleet of vehicles
 - choosing the best facilities to open or close

Network Analysis

- Direct path analysis : finding the **shortest path** between office and home
- Optimum routing: helping a pizza deliveryman visit numerous houses in the most time: efficient manner, that include length of the lines, their capacity, maximum travel rate and time



- Closest facility analysis : finding the closest hospital to an automobile accident

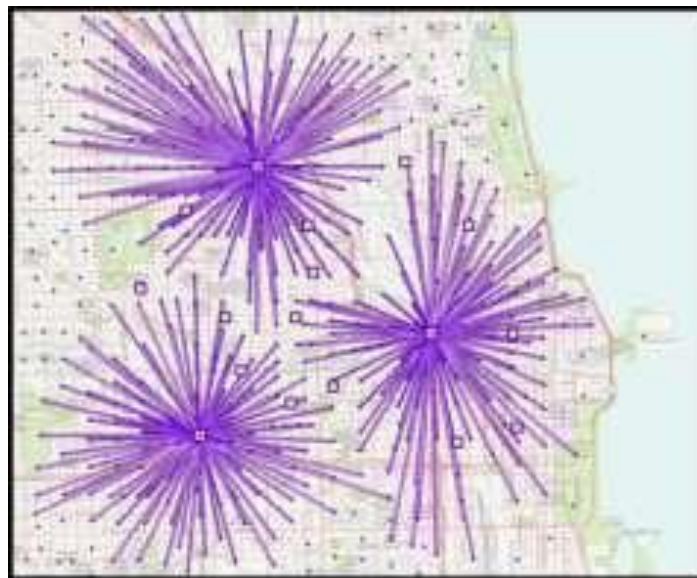


Network Analysis

Driving directions: the systems of computation also allow deriving directions



Service areas of Pizza shops to keep the Pizza hot enough



Drive time analysis: Helping a store to determine how many customers are within 5 km of driving

2. Topology Rules

Material of ArcGIS-ESRI

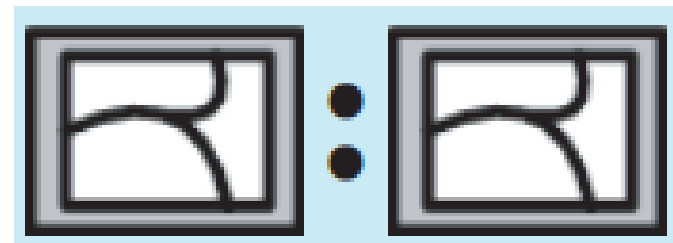
Topology allows:

- Model spatial **relationships** between objects or classes
- Define those relationships between **features in a single feature class** or subtype or **between two feature classes** or subtypes.
- Define the **spatial relationships** that meet the needs of data model.
- Topology errors are violations of the rules

- The topology rule occurs **within a single feature class (layer)** or subtype.

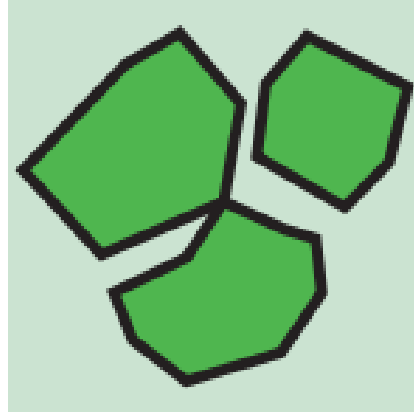


- The topology rule occurs between **two different feature class (layer)** or subtypes.

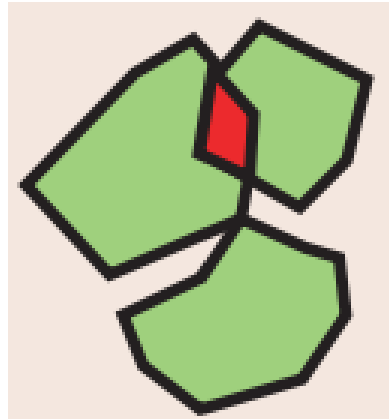


Polygons: Must not overlap

- Polygons must not overlap within a feature class or subtype. Polygons can be disconnected or touch at a point or touch along an edge



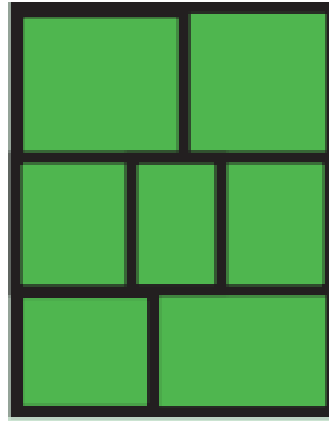
- Polygon errors are created from areas where polygons overlap.



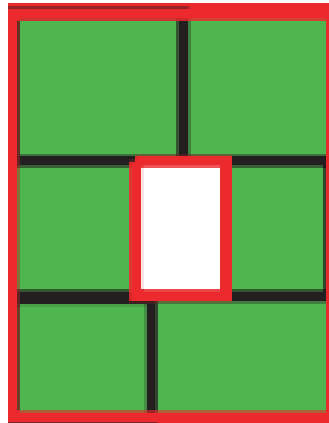
Ex. A voting district map cannot have any overlaps in its coverage

Polygons: Must not have gap

- Polygons must not have a void between them within a feature class or subtype.
- Line errors are created from the outlines of void areas in a single polygon or between polygons. Polygon boundaries that are not coincident with other polygon boundaries are errors.

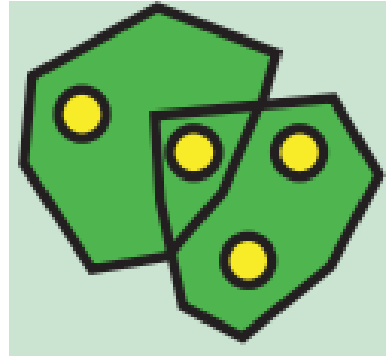


Ex: Soil polygons cannot include gaps or form voids—they must form a continuous fabric

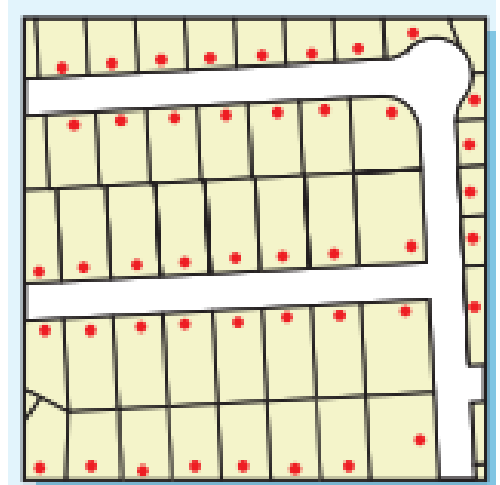


Polygons: Contains point

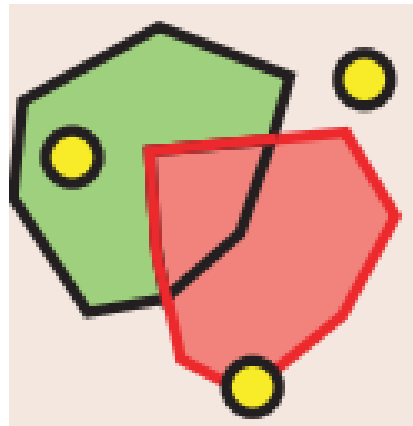
- Each polygon of the first feature class or subtype must contain within its boundaries at least one point of the second feature class or subtype.



Parcels must contain at least one address point.

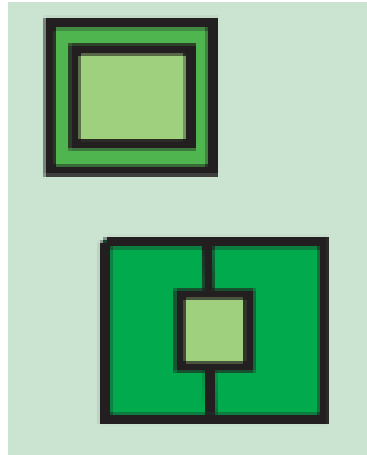


- Polygon errors are created from the polygons that do not contain at least one point. A point on the boundary of a polygon is not contained in that polygon.

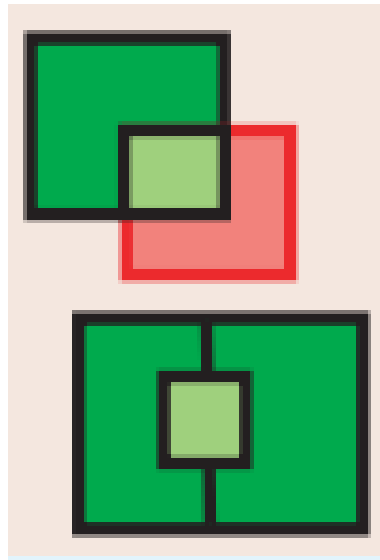


Polygons: Must be covered by feature class of

- The polygons in the first feature class or subtype must be covered by the polygons of the second feature class or subtype



- Polygon errors are created from the uncovered areas of the polygons in the first feature class or subtype.

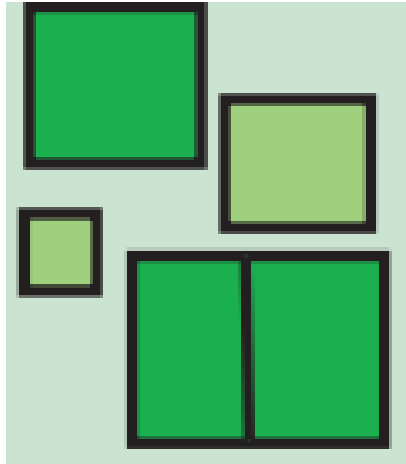


States are covered by counties



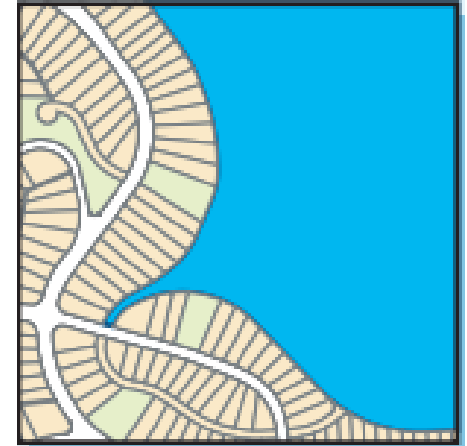
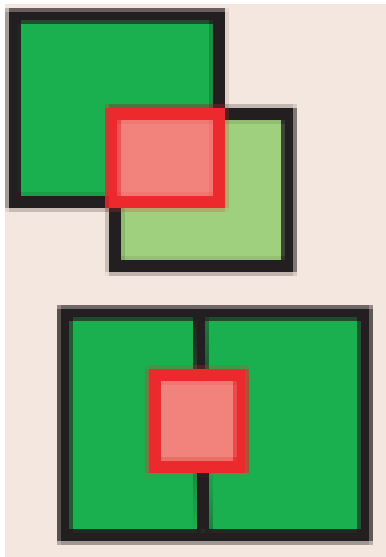
Polygons: Must not overlap with

- Polygons of the first feature class or subtype must not overlap polygons of the second feature class or subtype .



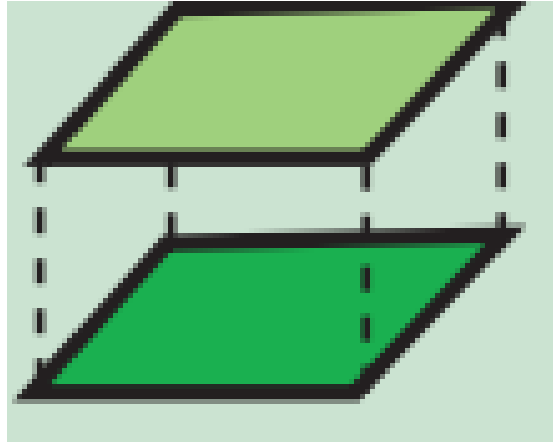
Lakes and land parcels from two different feature classes must not overlap.

- Polygon errors are created where polygons from the two feature classes or subtypes overlap

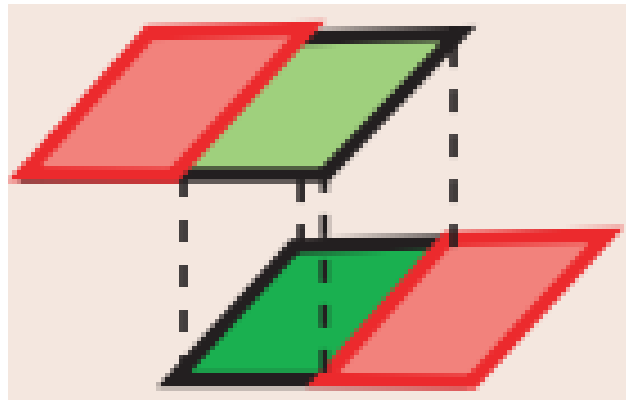
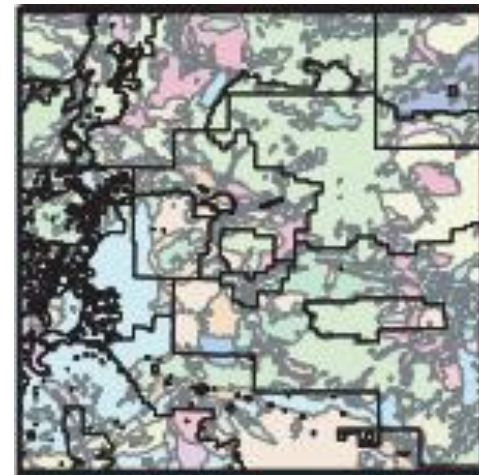


Polygons: Must cover each other

- All polygons in the first feature class and all polygons in the second feature class must cover each other.
 - FC1 Must be covered by feature class of FC2.
 - FC2 Must be covered by feature class of FC1.
- Polygon errors are created where any part of a polygon is not covered by one or more polygons in the other feature class or subtype

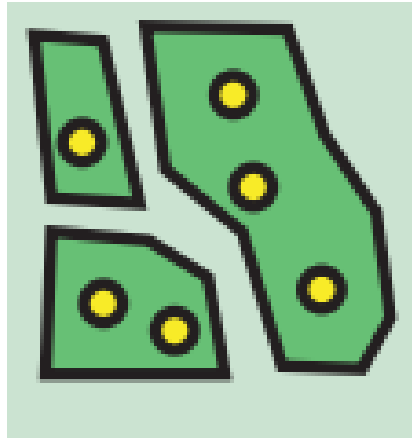


Vegetation and soils must cover each other

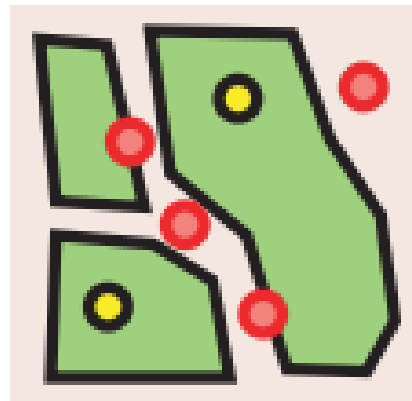


Point: Must be properly inside polygons

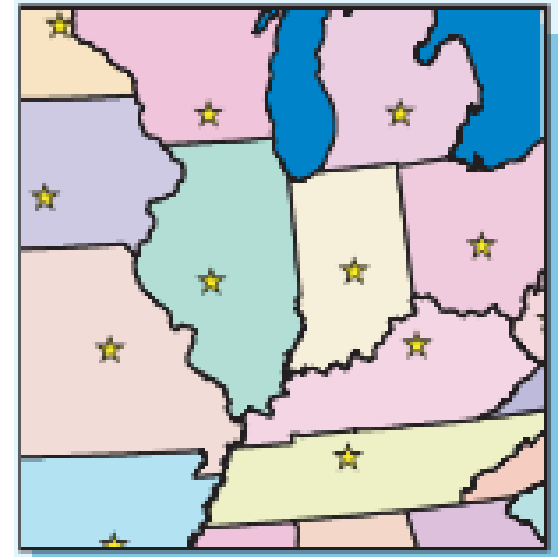
- Points in one feature class or subtype must be inside polygons of another feature class or subtype



- Point errors are created where the points are outside or touch the boundary of the polygons.

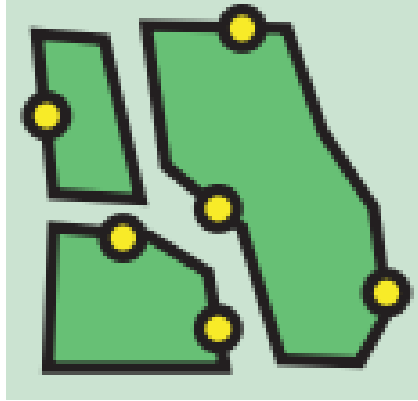


State capitals must be inside each state

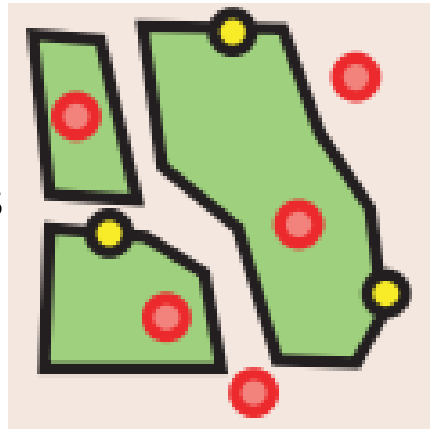


Point: Must be covered by boundary of

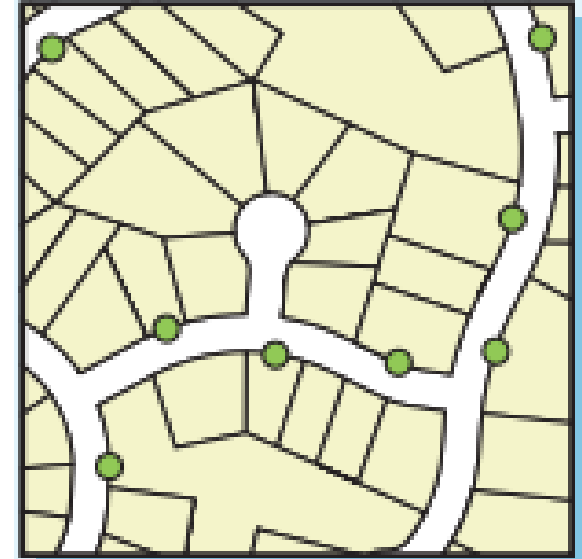
- Points in one feature class or subtype must touch boundaries of polygons from another feature class or subtype.



- Point errors are created where points do not touch the boundaries of polygons.

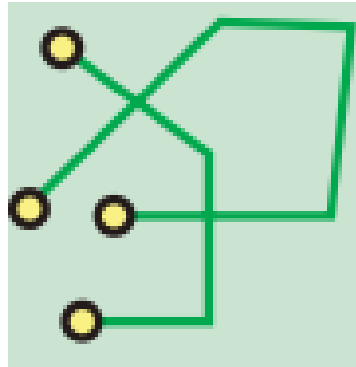


Utility service points might be required to be on the boundary of a parcel.



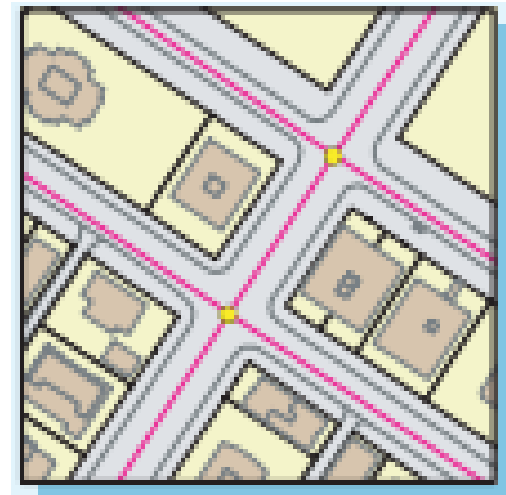
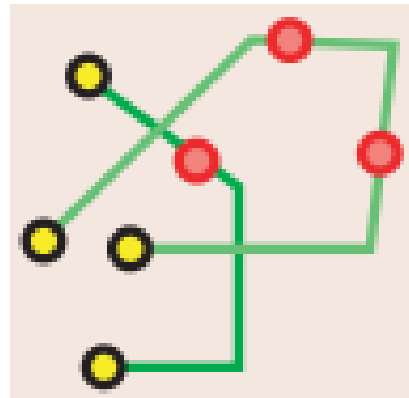
Point: Must be covered by endpoint of

- Points in one feature class or subtype must be covered by the ends of lines in another feature class or subtype.



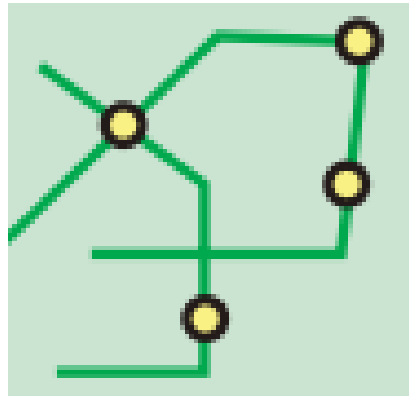
Street intersections must be covered by the endpoints of street centerlines

- Point errors are created on the points that are not covered by the ends of lines

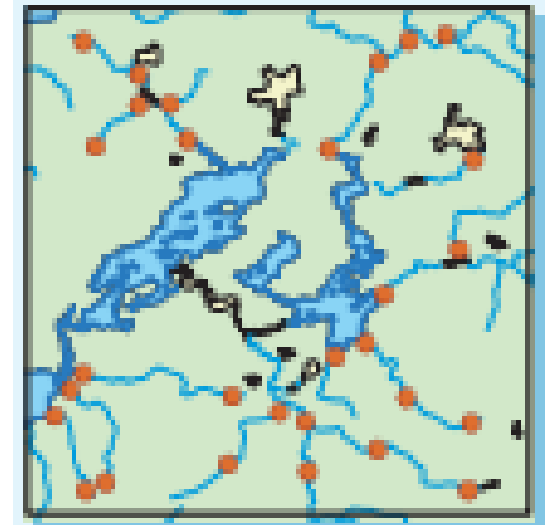


Point: Point must be covered by line

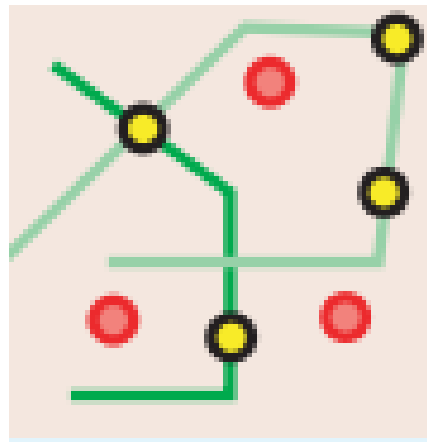
- Points in one feature class or subtype must be covered by lines in another feature class or subtype



Monitoring stations must fall along streams

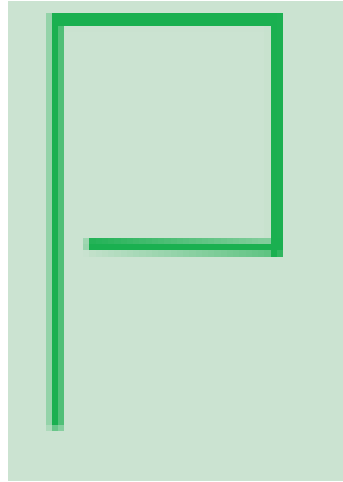


- Point errors are created on the points that are not covered by lines



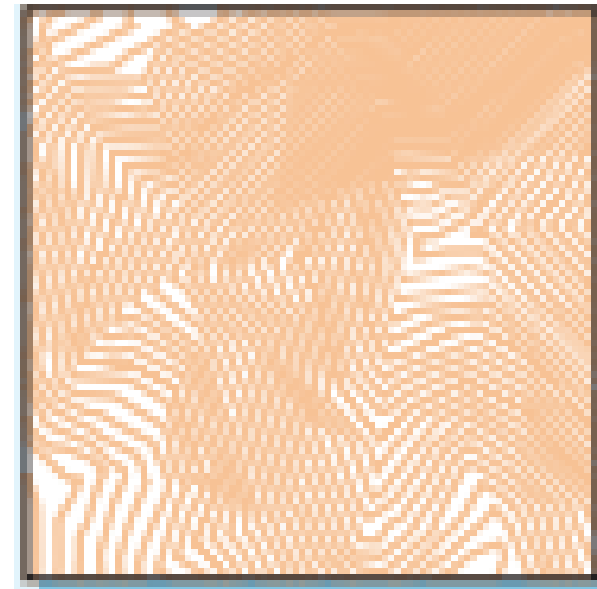
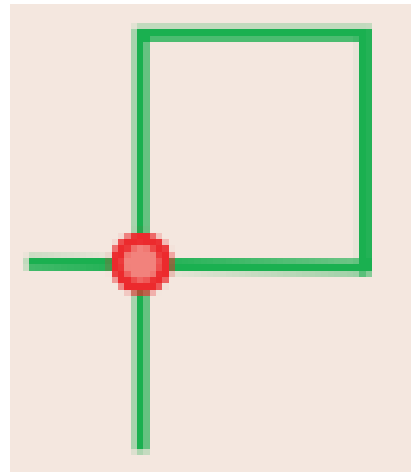
Line: Must not self intersect

- Lines must not cross or overlap themselves within a feature class or subtype. Lines can touch themselves and touch, intersect, and overlap other lines



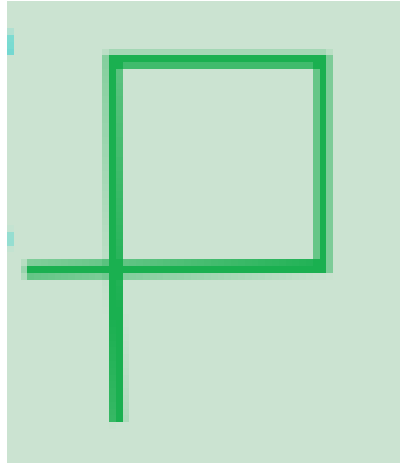
Contour lines cannot intersect themselves

- Line errors are created where lines overlap themselves, and point errors are created where lines cross themselves

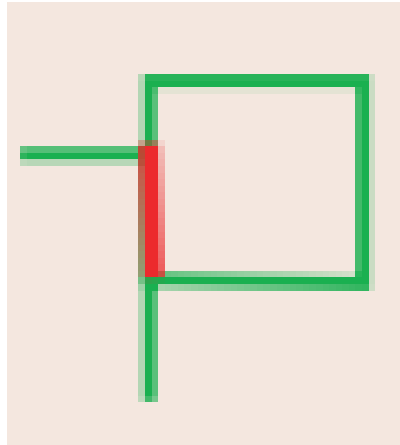


Line: Must not self overlap

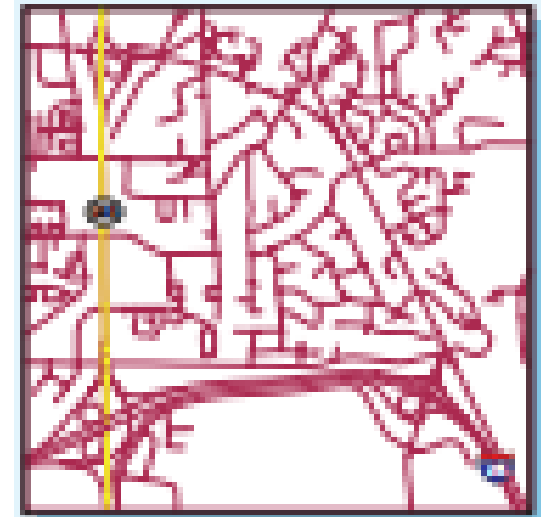
- Lines must not overlap themselves within a feature class or subtype. Lines can touch, intersect, and overlap lines in another feature class or subtype.



- Line errors are created where lines overlap themselves.



For transportation analysis, street and highway segments of the same feature should not overlap themselves



3. Buffers, Overlays, and Spatial Joins

- Buffers
- Overlays
 - Clip
 - Intersection
 - Union
 - Dissolve
 - Merge
- Spatial Joins

Buffers

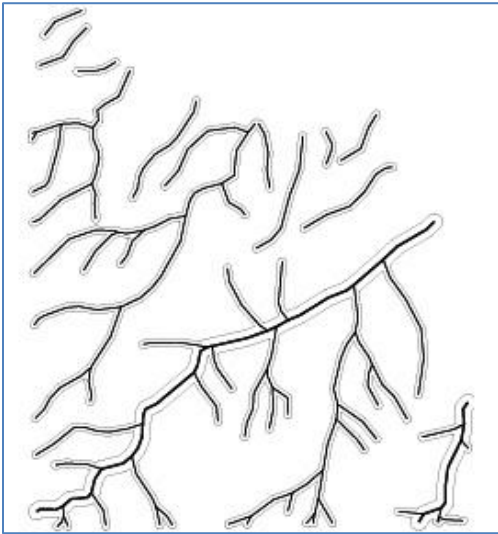
Buffering separates features into two areas: one area that is within a specified distance of selected features and the other area that is beyond. The area that is within the specified distance is called the buffer zone.

- Can add a buffer to a point, line, or polygon
- Attention: map units when specifying buffer distance



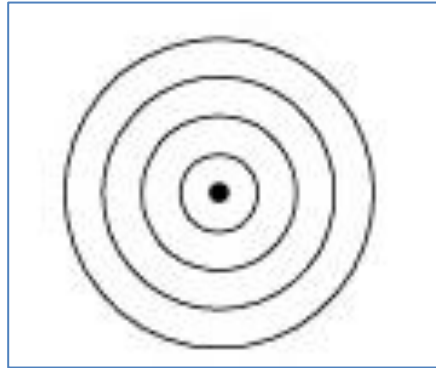
Variations in Buffering

1. The buffer distance can vary according to the values of a given field.



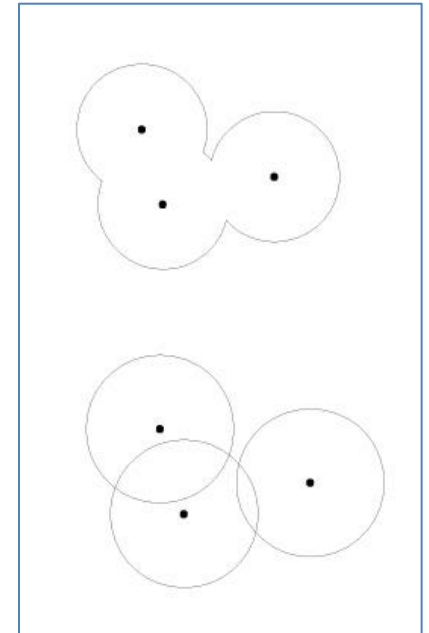
Buffering with different buffer distances

2. Features may have more than one buffer zone.



A nuclear power plant may be buffered with distances of 5, 10, 15, and 20 miles, thus forming multiple rings around the plant **(Why?)**

3. Boundaries of buffer zones may remain separate polygons or be dissolved (no overlapped areas)



Buffer zones dissolved (top) or not dissolved (bottom) **(Examples?)**

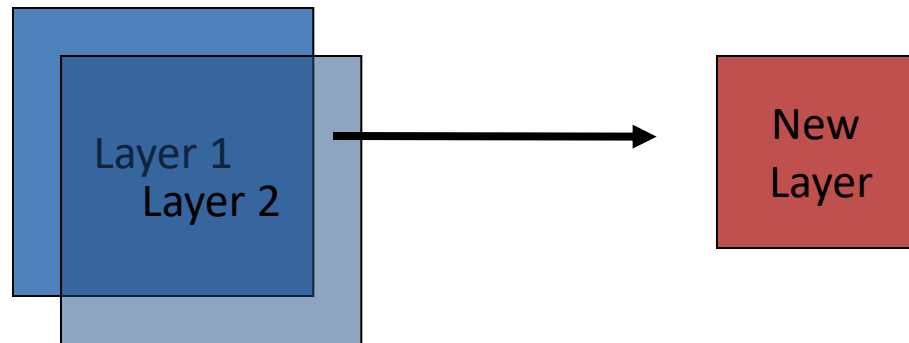
Applications of Buffering

- Ex 1. Regulation of a city: no liquor stores or pornographic shops shall be within 1000 feet of a school or a church.
- Ex 2. Government regulations: logging operations must be at least 2 miles away from any stream to minimize the sedimentation problem and set the 2-mile buffer zones of streams as the exclusion zones.
- Ex 3. Regulation: No oil and gas well drilling within 500 feet of roads or highways; within 200 feet of trails; within 500 feet of streams, lakes, ponds, or reservoirs; or within 400 feet of springs.
- Ex 4. A dumping ground must 1 km far from the village, 1.5 km far from the school and 2 km far from water resources (How to find the location for the dumping ground?)

What is an overlay?

Overlay combines the geometries and attributes of two features to create the output. One of the two feature layers is called the input layer and the other the overlay layer.

- Take two or more layers
 - Typically with the same extent (they are one on top of another)
- Combine the layers in some way
 - Specific overlay techniques
- Create an output layer



What are the criteria to perform overlay?

All layer must be the same coordinate systems to avoid crashing the program or inaccurate results

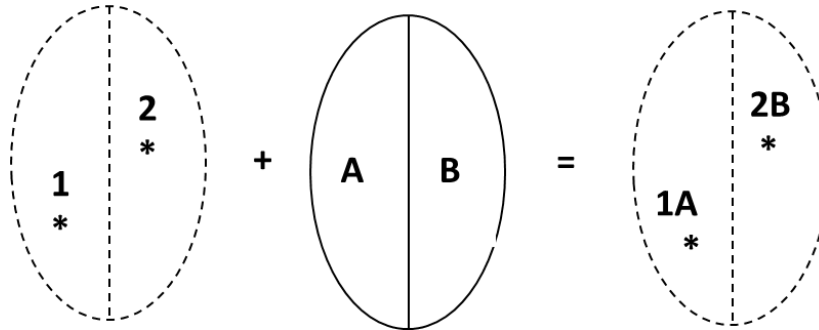
Feature Type and Overlay

Point in Polygon

Line in Polygon

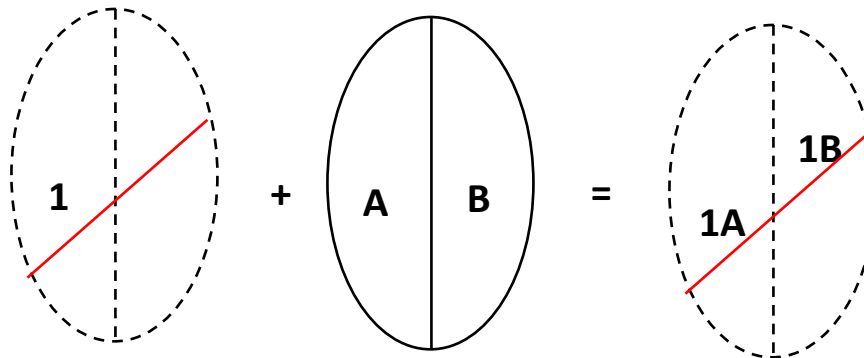
Polygon on Polygon

Point in polygon overlay



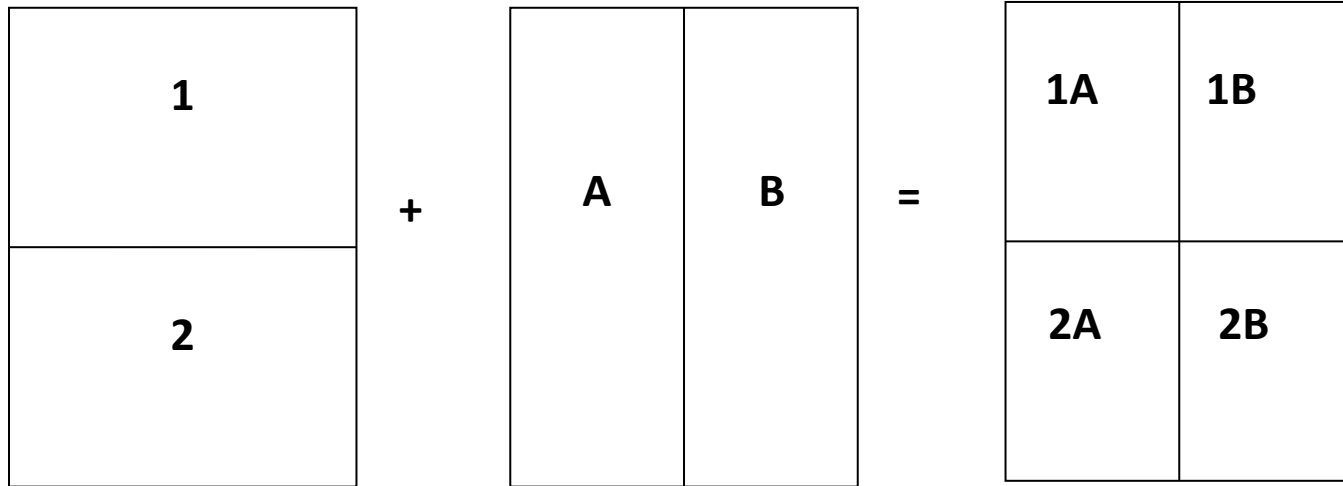
*The input is a point feature (the dashed lines are not part of the point).
The output is also a point, which has attribute data from the overlay
polygon map.*

Line in polygon overlay



*The input is a line (the dashed lines not part of the line). The output is also a line
map. But the output differs from the: the line is broken into two segments, and
the line segments have attribute data from the overlay polygon.*

Polygon-on-polygon overlay



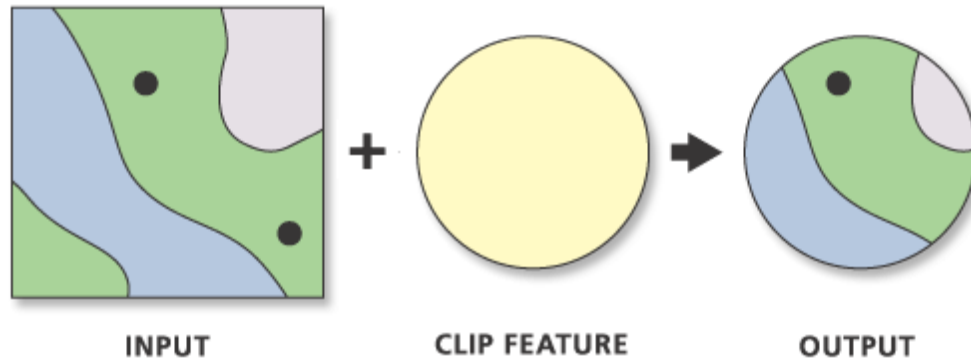
The two maps to be overlaid have the same area extent. The output combines the geometry, and attribute data from the two maps are combined into every single polygon.

Overlaying Vector Data

- Common vector overlay/geoprocessing operations
 - Clip (cookie cutter)
 - Intersect – keeps only common features between two layers
 - Union – combines all features in two layers
 - Dissolve – combine (simplify) features in one layer
 - Merge – combine two layers into one
- Always recalculate areas/perimeters of polygons after one of these operations

Clip

- Use one layer as a boundary to clip (cut) out features from another layer



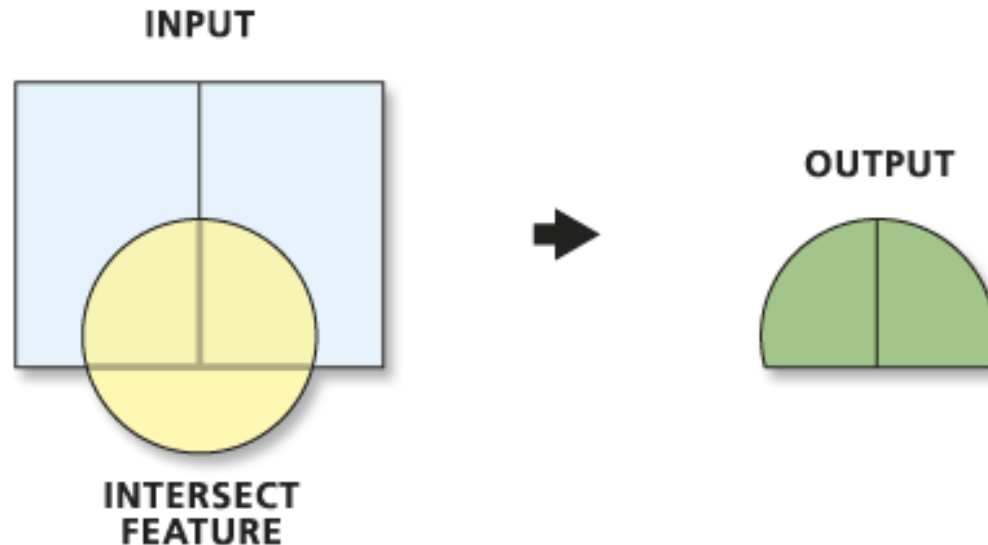
Be sure to have correct layers as the input layer (the one to be cut) and the clip layer (the one to use as a cutting guide)

Must use polygons as the clip layer

Can use selected features or all features as a clip layer

Intersect

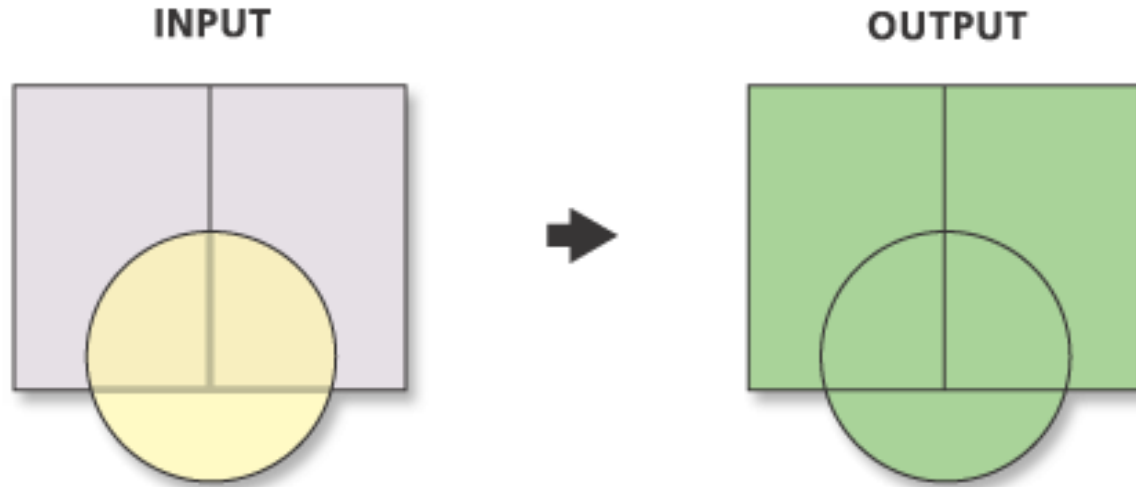
- Combines two layers and outputs only the features common (spatially) to both layers



- Be sure to know which layer is the input layer and which is the overlay layer
- The overlay layer must be polygons
- Attributes will come from **both layers**
- Will need to **re-calculate** polygon areas

Union

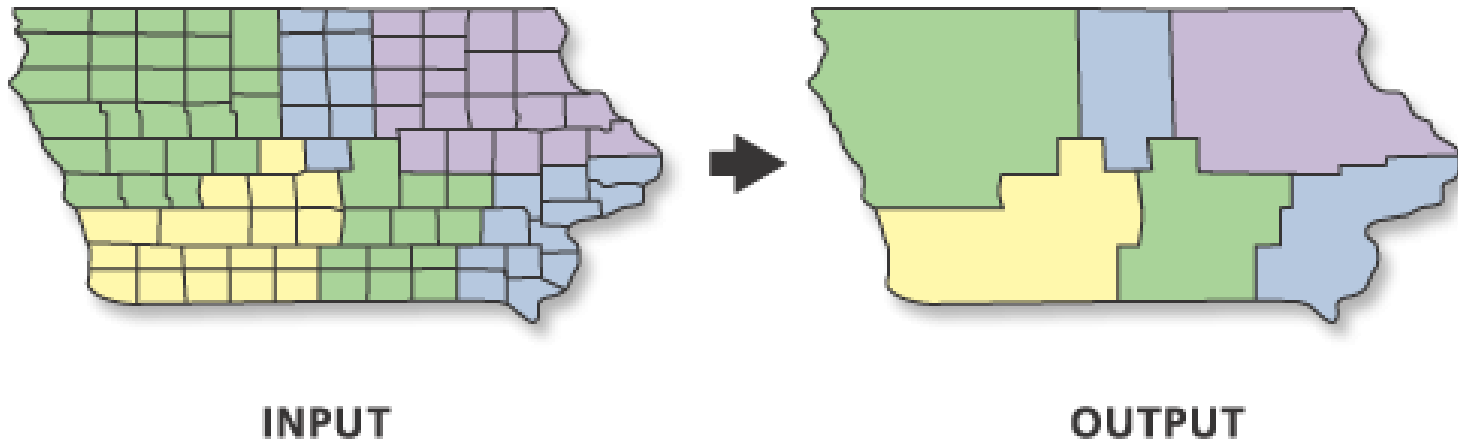
- Combines two layers and all features (whether or not they overlap)



- Be sure to know which is the input and which is the overlay layer
- May use all or selected features as overlay
- Overlay must be polygon
- Attributes will come from **both layers**

Dissolve

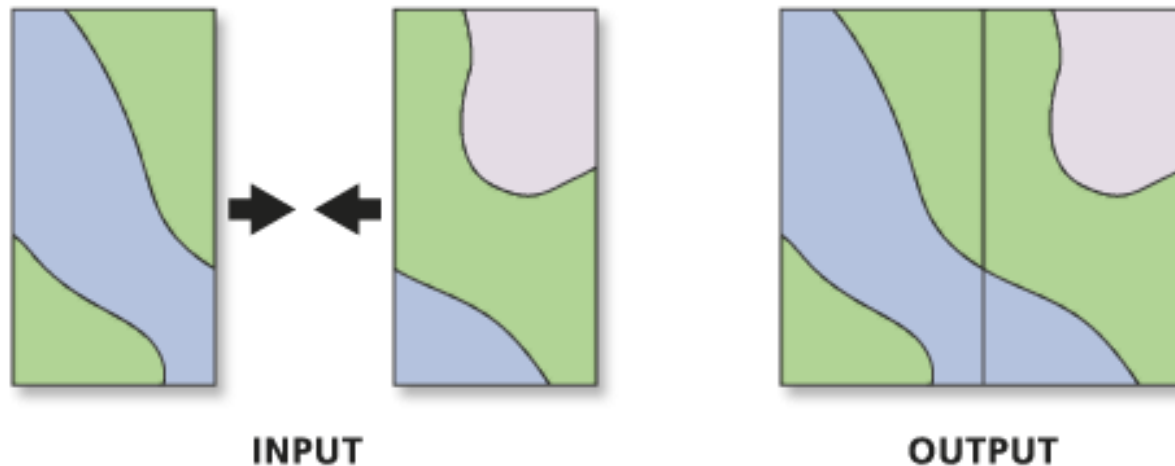
- Combine features in a layer based on one attribute
- Example: aggregate counties in US by state name



- Have to specify **common feature**
- Points, lines, and polygons
- Need to specify how to deal with remaining attributes
 - First or last
 - Sum, min, max, mean

Merge

- Combine features from two or more layers into one layer



- Can have multiple layers
- Must specify which layer will define **attribute fields** for the output
- Need to be aware of:
 - Differing attribute tables
 - Presence of gaps (spatially) in merged layers

Summary: vector overlay

- Powerful tools for creating new information
- Consider attributes
- Recalculate area and perimeter
- Files with **different coordinate systems** can cause problems – convert coordinate systems first

4/ Raster analysis

Major material by University of Twente

4.1 Measurements on raster

4.2 Classification

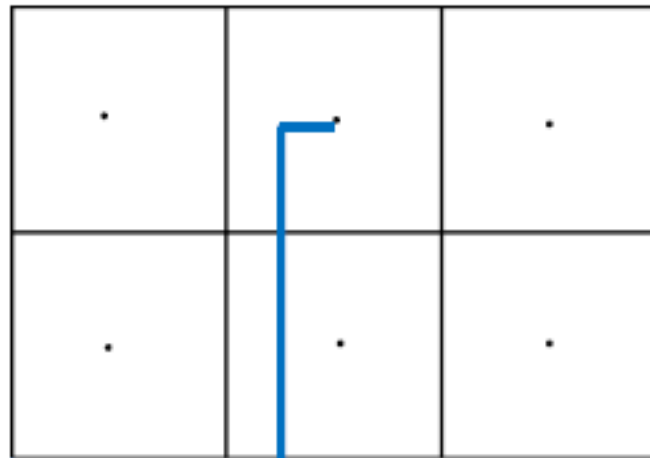
4.3 Overlay raster

4.4 Surface modeling operation

4.1 Measurements on raster

Location

Resolution: 20 x 20 meters

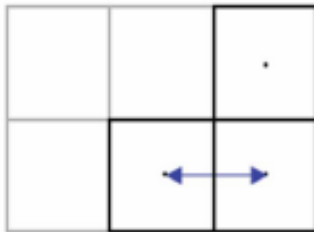


- Location of an individual cell is derived from **anchor point** and **resolution**
- A cell's location can be defined by the coordinates of its *lower left corner* or its *midpoint*

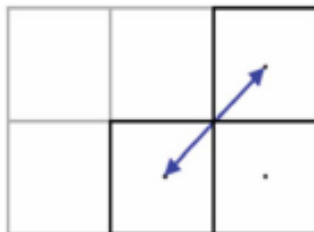
Distance

- The Euclidean distance between two cells is the distance between their mid-points

Cell size = 30 m x 30 m

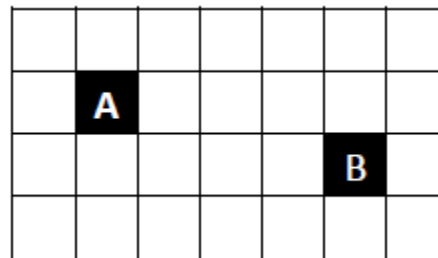
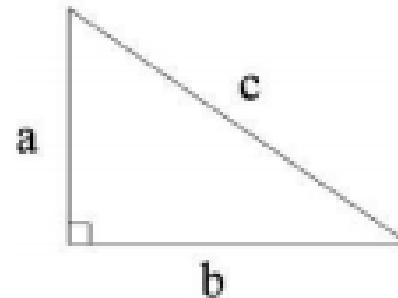


Distance 30 m



Distance $30 * \sqrt{2}$

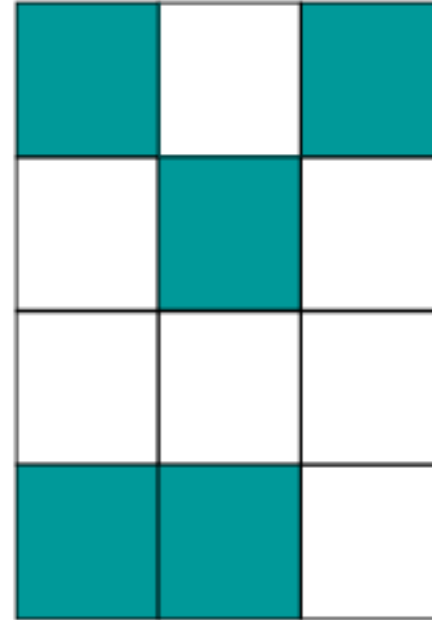
$$a^2 + b^2 = c^2$$



$$\sqrt{(2-6)^2 + (2-3)^2} * \text{cell size} \approx 4.1 * \text{cell size}$$

Area

- **Area Size =**
number of cells * cell size
- The number of cells is also called the **frequency** or **count**

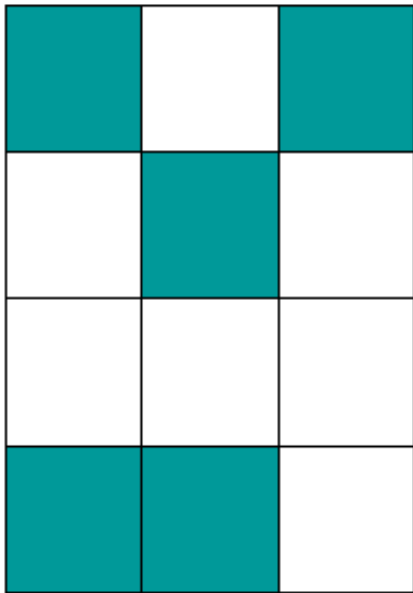


Cell size: 30 m x 30 m

$$900 * 5 = 4500 \text{ m}^2$$

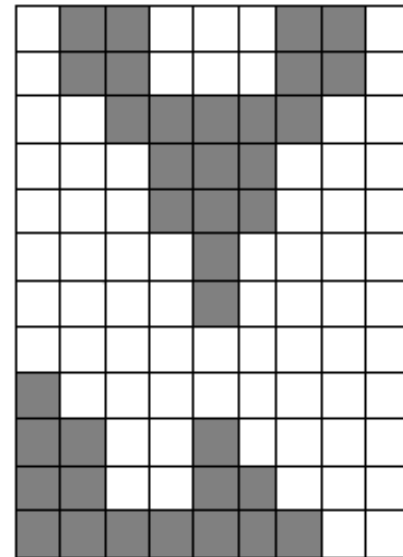
Area

- Does the area depend on the raster size?



Cell size: 30 m x 30 m

$$900 * 5 = 4500 \text{ m}^2$$



Cell size: 10 m x 10 m

$$100 * 36 = 3600 \text{ m}^2$$

4.2 Raster classification

- it removes details to reveal patterns
- It produces a new output dataset
 - a new attribute in vector
 - a new raster layer in raster
- If input = classification → reclassification
- User controlled classification
 - Classification table
- Automatic classification
 - Equal interval technique
 - Equal frequency technique

(Similar to vector classification)

Raster classification

- In user-controlled classification we indicate the classification **attribute** (“old” and “new”)
- This is normally done via a **classification table**

Old value	New value
391 - 2474	1
2475 - 6030	2
6031 - 8164	3

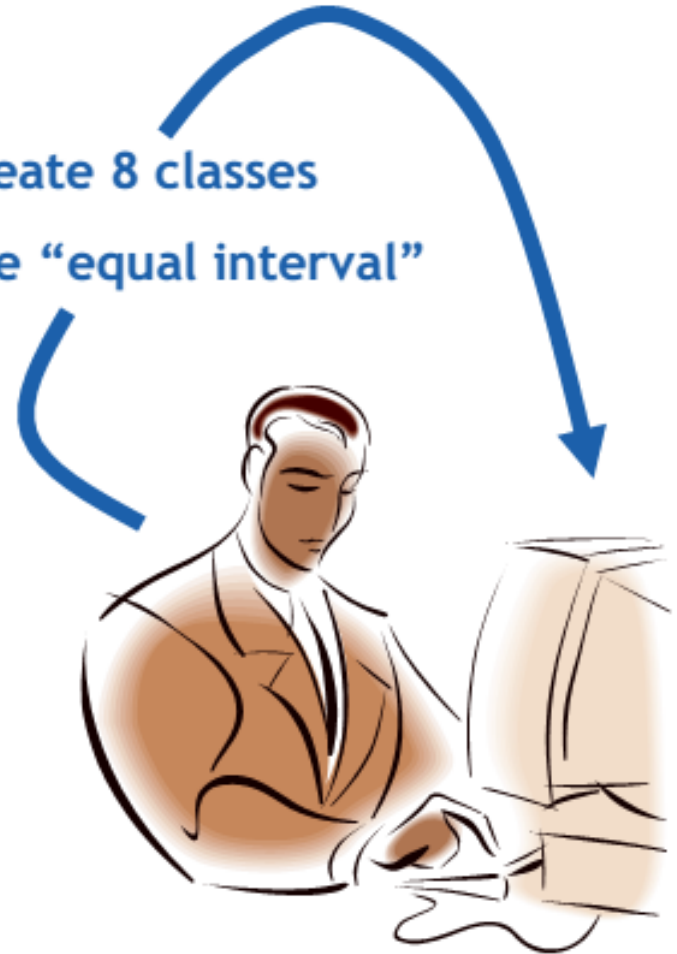
Code	Old value	New value
10	Planned Residential	Residential
20	Industrial	Commercial
30	Commercial	Commercial

Top table, the original values are ranges; lower table the original values were a classification.

CLASSIFICATION

- **User** specifies the number of output classes.
- **Computer** decides the class break points.
- Two possibilities are discussed:
 - *Equal frequency*
 - *Equal interval*

Create 8 classes
Use “equal interval”



CLASSIFICATION

Equal interval

- calculated as $(v_{max} - v_{min}) / n$
- v_{max} is the maximum attribute value, v_{min} is the minimum attribute value and n is the number of classes.

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

(a) original raster

1	1	1	1	4
2	2	3	2	5
2	2	2	1	5
2	3	3	4	4
2	1	1	1	1

(b) equal interval classification

- In our example: $(10-1) / 5 \approx 2$
- Each class “contains” two values

Two original values per class

original value	new value	# cells
1,2	1	9
3,4	2	8
5,6	3	3
7,8	4	3
9,10	5	2

CLASSIFICATION

Equal frequency

- Also called **quantile**
- calculated as

$$\frac{\text{total number of features}}{\text{number of classes (n)}}$$

- The objective is to create categories with roughly equal number of features (or cells).

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

(a) original raster

1	1	1	2	5
3	3	4	3	5
3	2	2	2	5
3	4	4	5	5
3	2	1	1	1

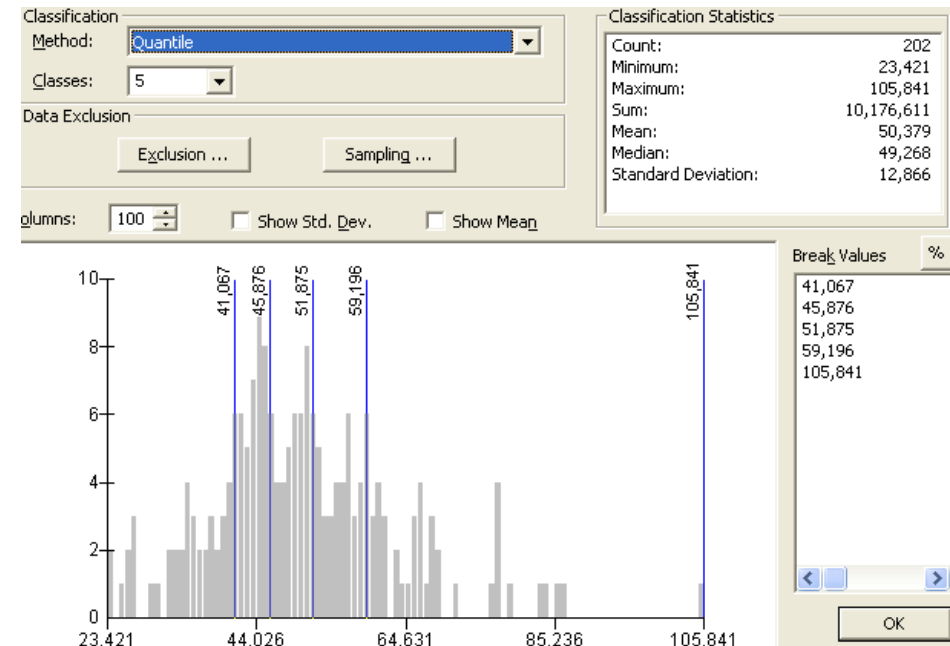
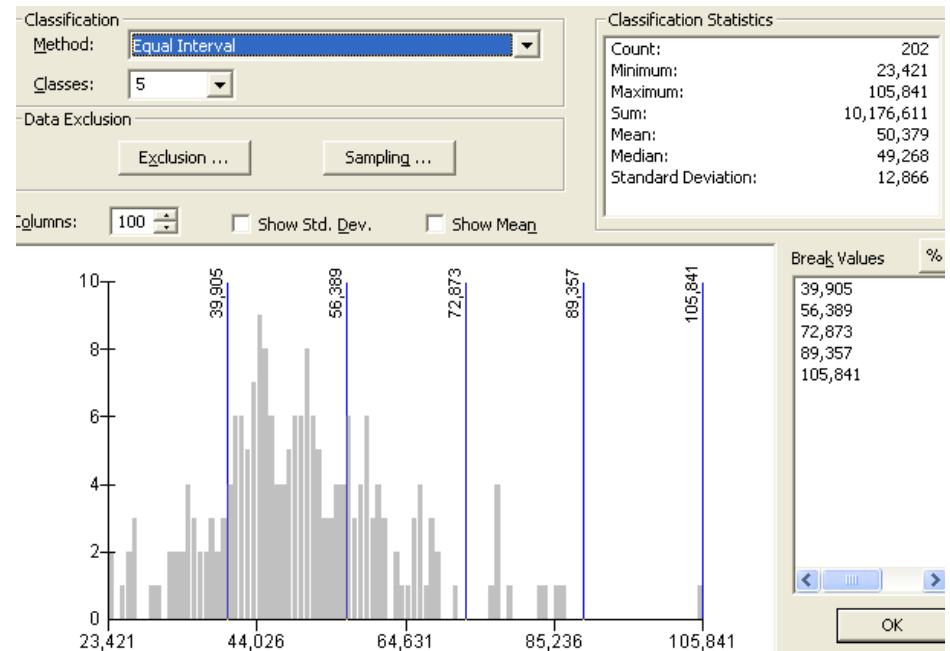
(c) equal frequency classification

Total number of cells $25 / 5 = 5$ cells per class

original value	new value	# cells
1	1	6
2,3	2	5
4	3	6
5,6	4	3
8,9,10	5	5

CLASSIFICATION

- Horizontal axis = original values
- Vertical axis = frequency
- Class boundaries are the blue vertical lines
- The **equal interval** example; the blue lines are equally spaced
- The **equal frequency technique**; As the frequency goes up, the class intervals become narrower



Equal classification

Advantages:

- Easy to understand, intuitive appeal
- Each class represents an equal range or amount of the original data range.
- Good for rectangular data distributions

Disadvantages:

- Not good for skewed data distributions—nearly all values appear in one class.

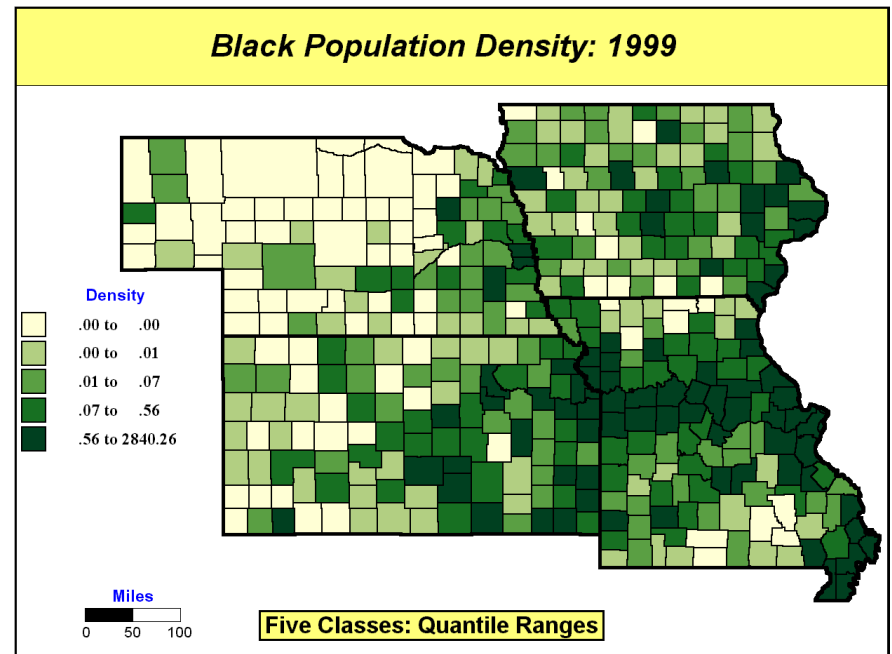
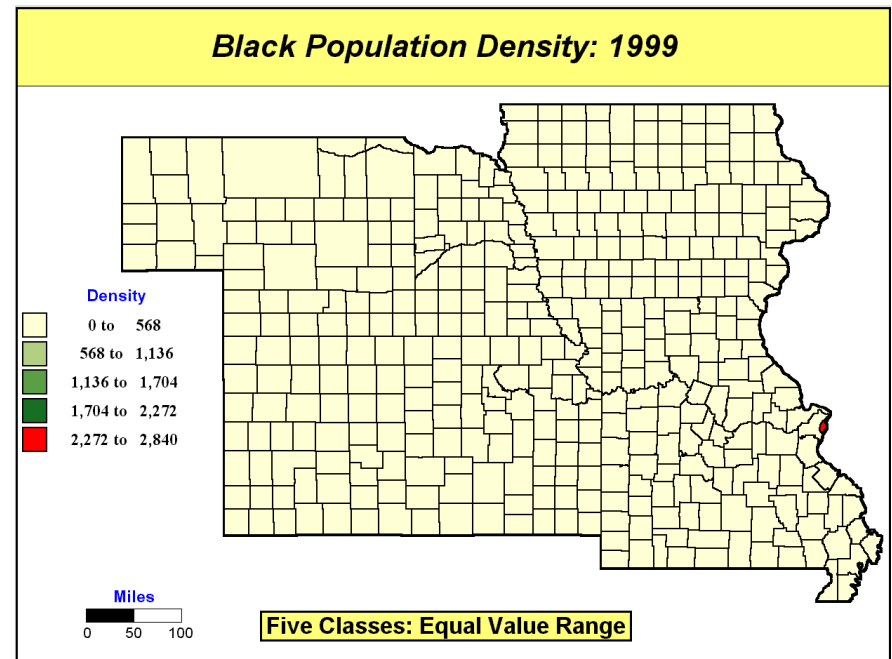
Quantile classification

Advantages:

- Each class has equal representation on the map.
- Intuitive appeal: map readers like to be able to identify the “top 20%” or the “bottom 20%”

Disadvantages:

- Very irregular break points unless data have rectangular distribution.
- Break points often seem arbitrary.

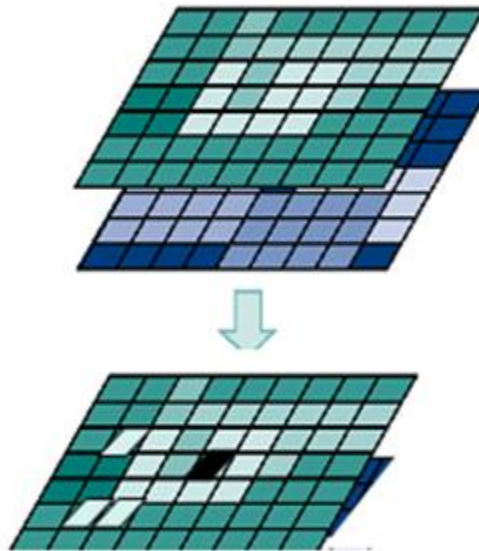


4.3 Overlay

- Arithmetic operators
- Comparison operators
- Logical operators
- Conditional expressions
- Decision tables

RASTER OVERLAY

- New cell values are calculated using **map algebra / raster calculus**.
- Performed on *cell-by-cell* basis.
- *No geometric* calculation

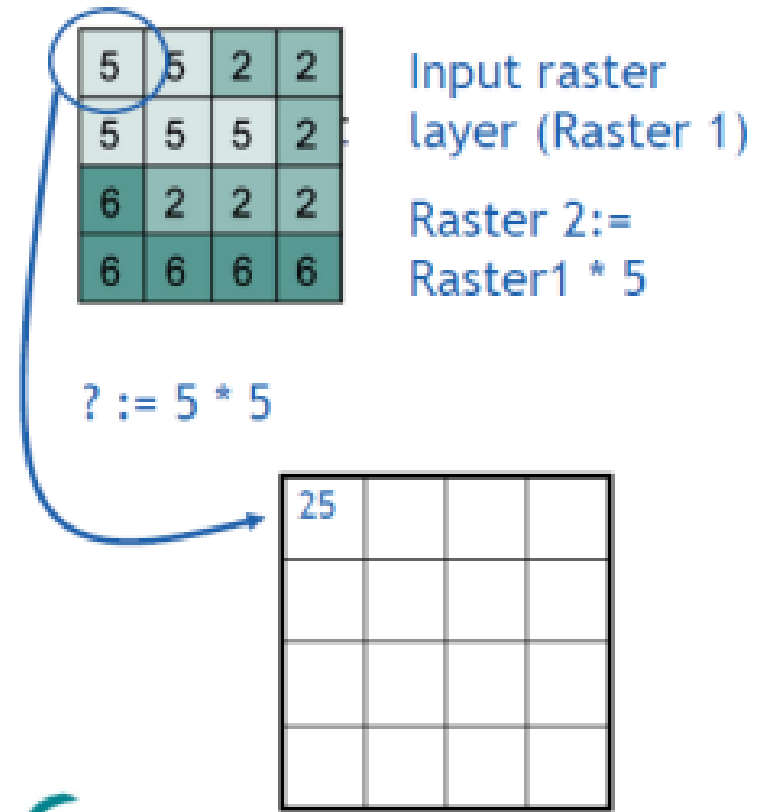


Output_raster_name := Raster_calculus_expression

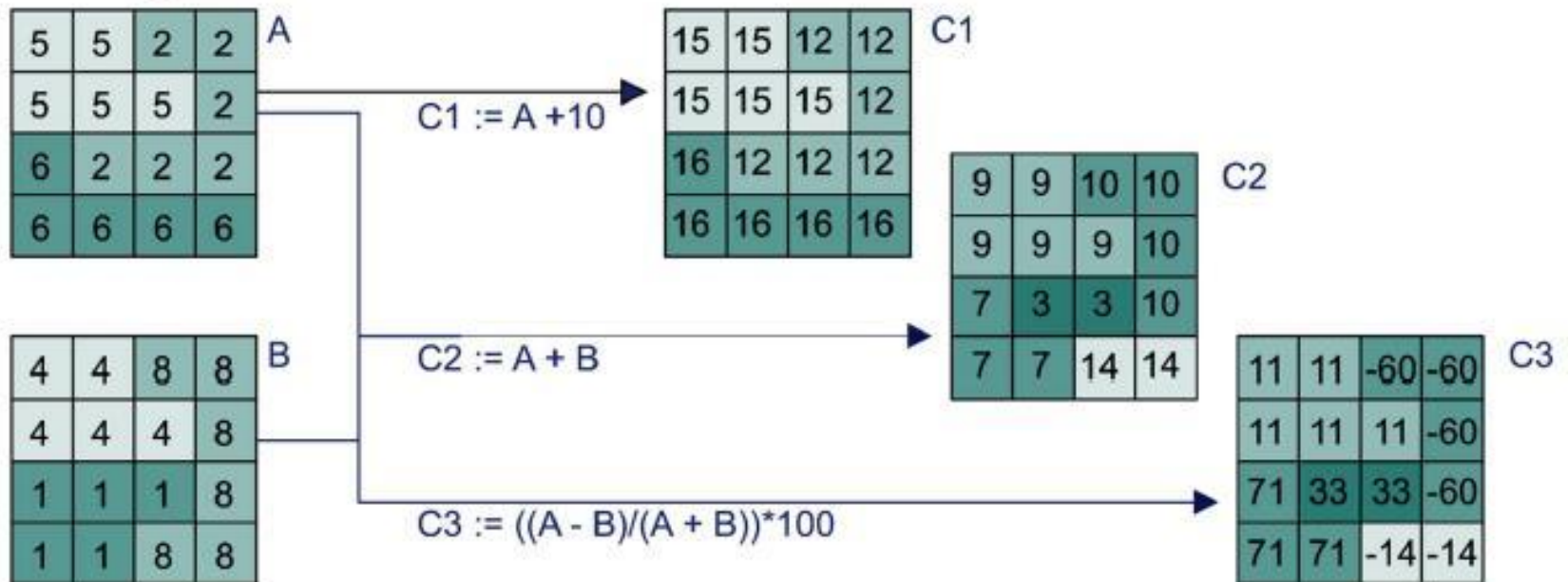
Overlay – Arithmetic Operators

Arithmetic operators

- $+$, $-$, $*$, $/$
- MOD (modulo division)
- DIV (integer division)
- Goniometric operators
sin, cos, tan, asin, acos, atan.
- Etc.



Overlay – Arithmetic Operators



What are criteria to do arithmetic operation?

Comparison Operators

Comparison operators: <, <=, =, >=, >, <>

- test whether one expression is larger, smaller, equal, etc. than another expression
- can be used in combination with logical operators or conditional functions
- Using only a comparison operator in a formula, the formula will be a Boolean statement, e.g. MapC := MapA <> MapB

Comparison Operators



25	25	15	10
15	20	20	10
10	15	20	5
5	10	15	5

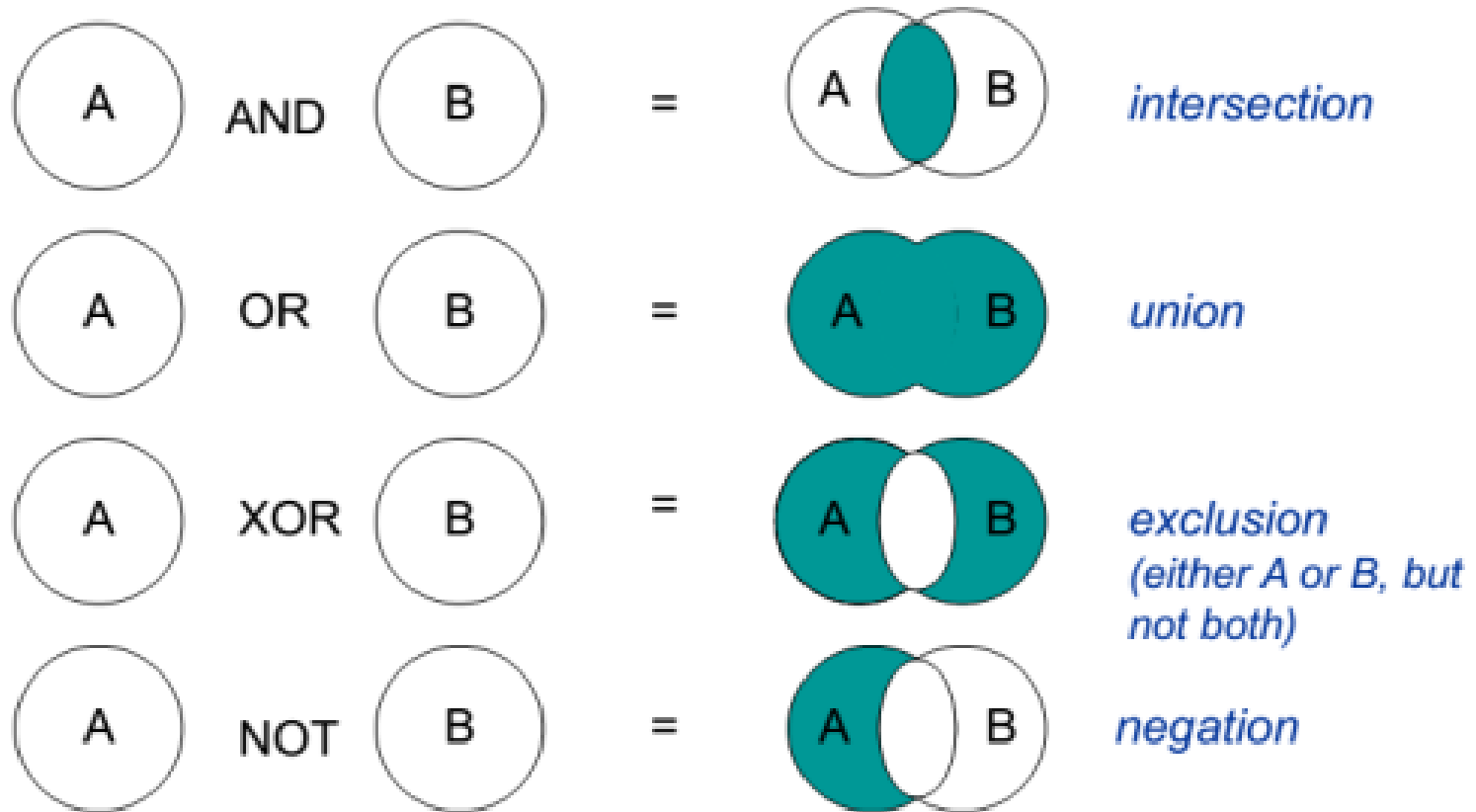
Output_raster := Input_raster < 15

True: 1

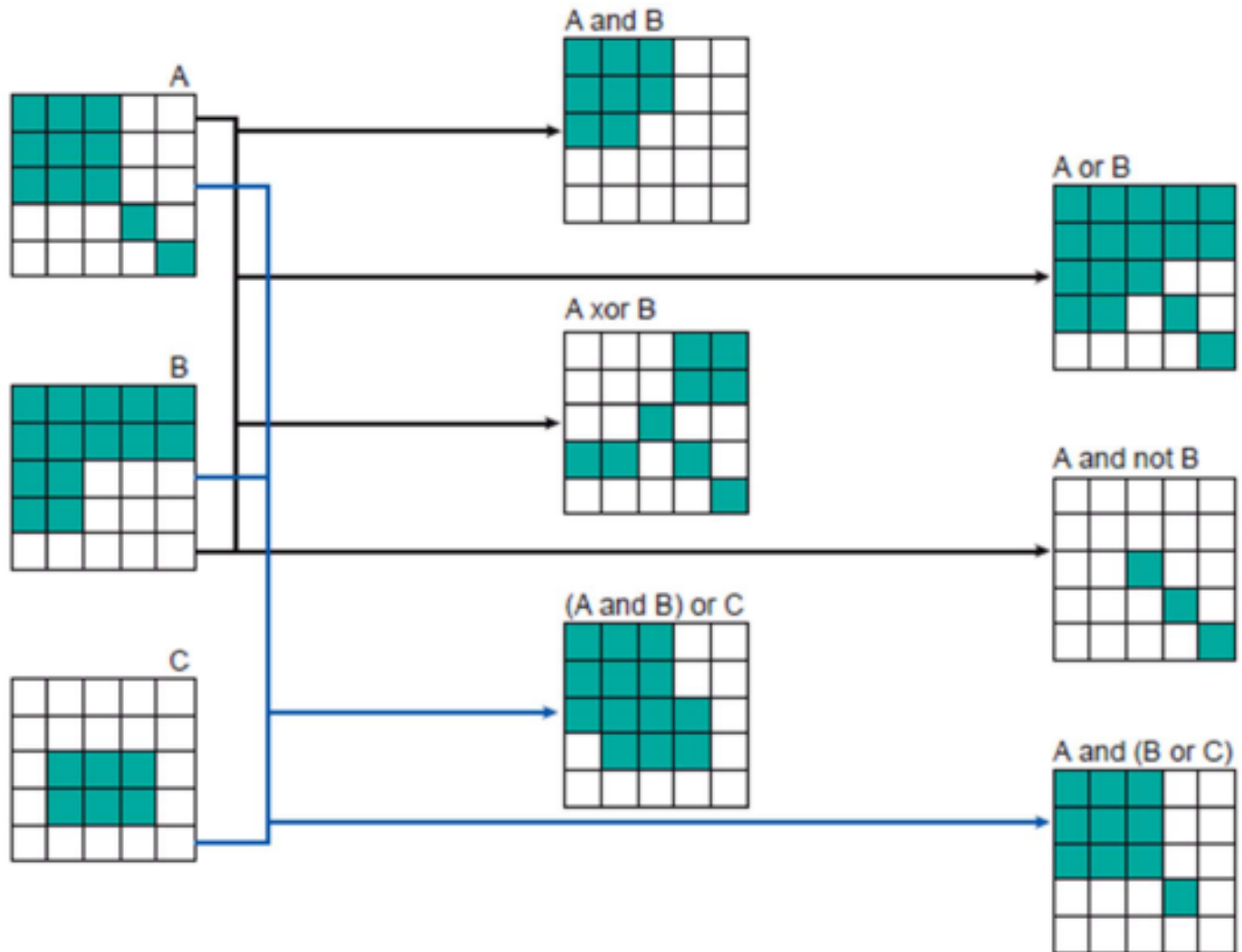
False: 0

0	0	0	1
0	0	0	1
1	0	0	1
1	1	0	1

Logical Operators



Logical Operators

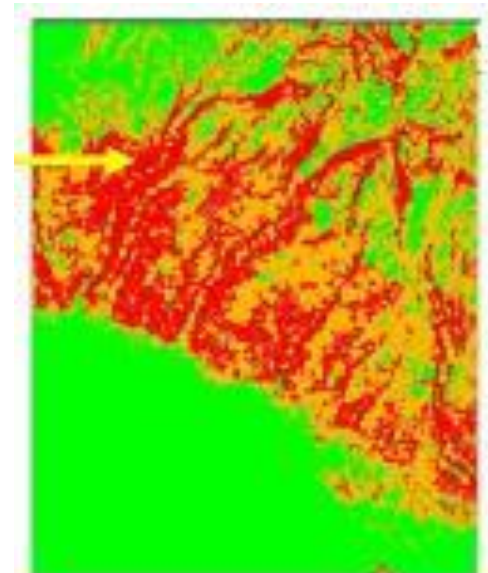
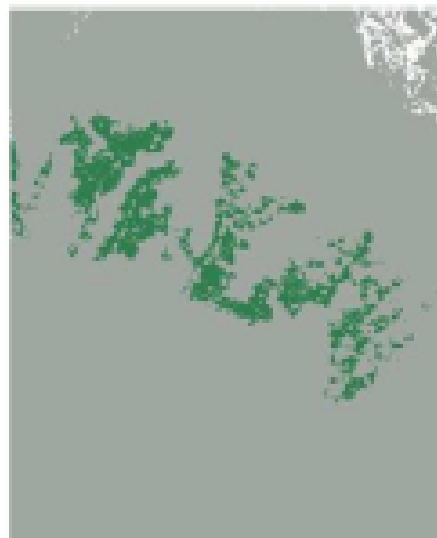


Logical Operators



Landuse = forest

Steep slope
Forest
AND



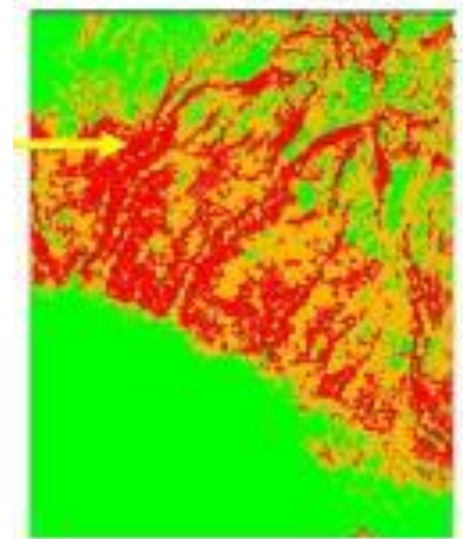
Slope class = steep

Logical Operators



Landuse = forest

Steep slope
Forest
OR



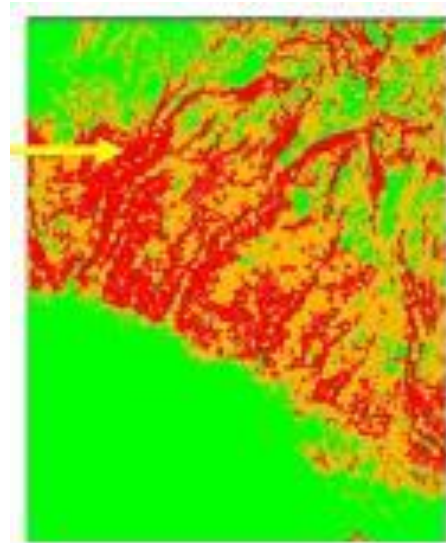
Slope class = steep

Logical Operators



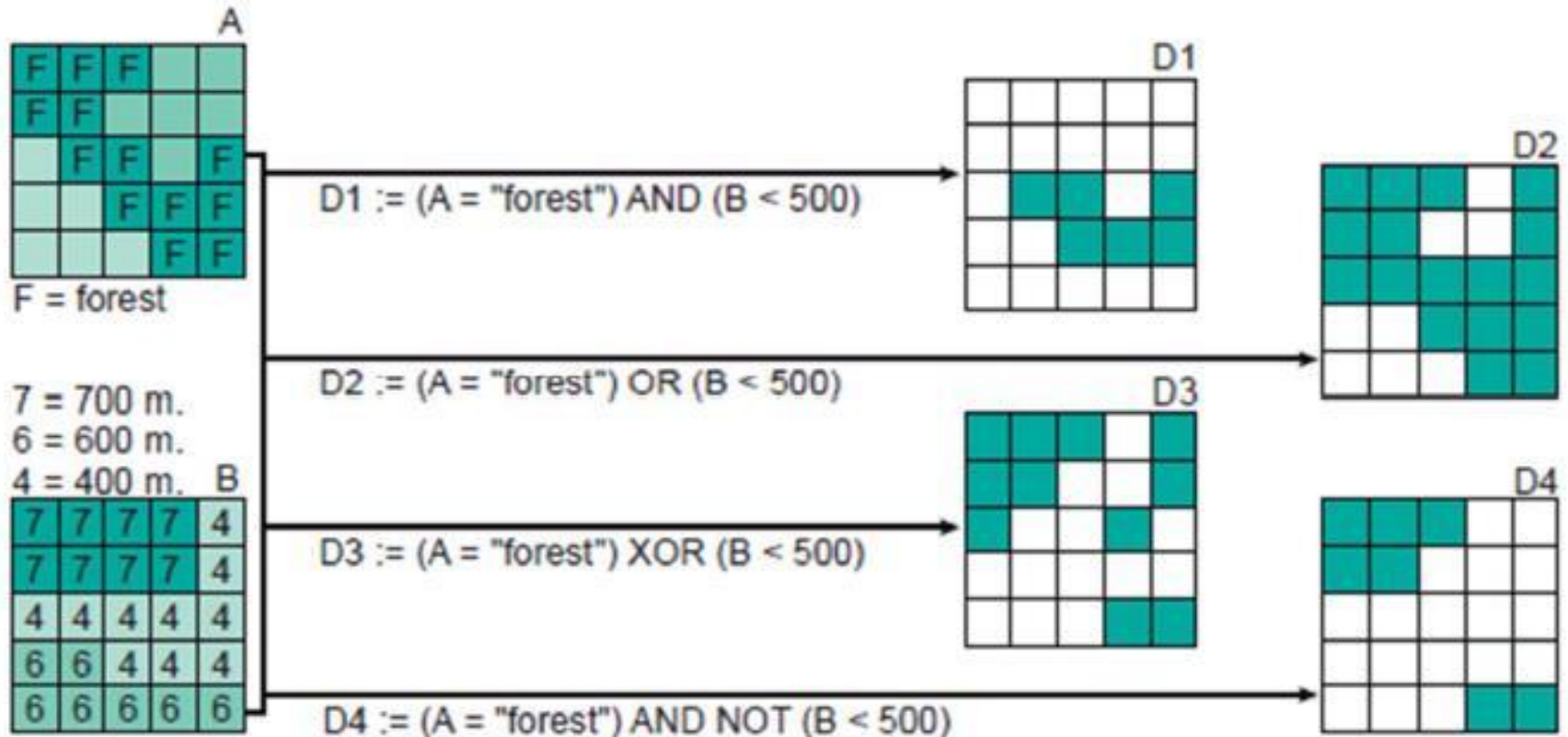
Landuse = forest

Steep slope
Forest
XOR



Slope class = steep

Comparison & Logical Operators



Conditional Operators

Conditional expressions:

- test a *condition*
- if the expression is **true**, the *then_expression* will be evaluated
- if the expression is **false** the *else_expression* will be executed.
- A conditional expression can lead to many different types of outputs, numeric value but also true/false

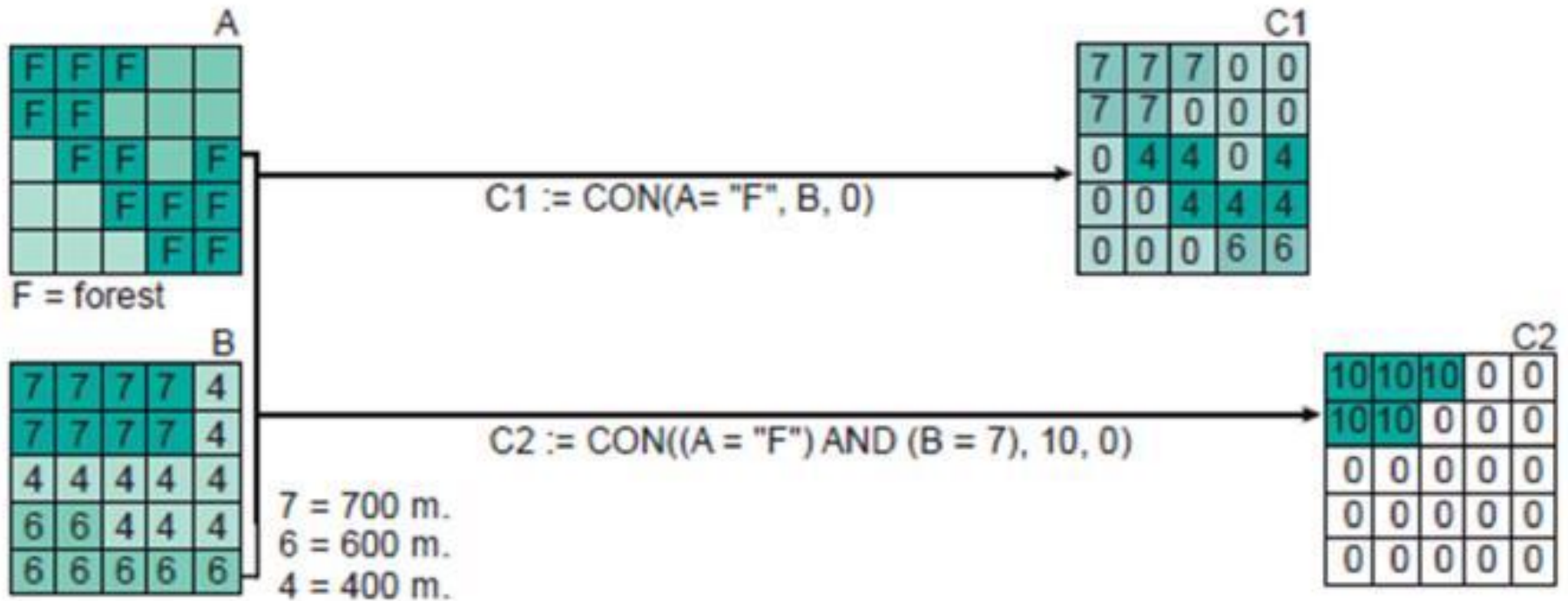
A				B			
25	25	15	10	4	4	4	2
15	20	20	10	2	6	4	2
10	15	20	5	2	6	6	4
5	10	15	5	2	2	2	5

Output:=Con (A <15 OR B=2, A, 40)

40	40	40	10
15	40	40	10
10	40	40	5
5	10	15	5

Output_raster:= CON (condition, then_expression, else_expression)

Conditional Operators

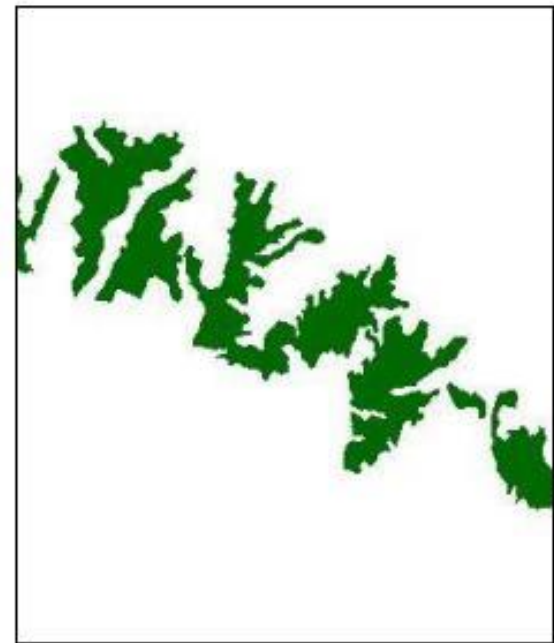


Conditional Operators

Show only the areas covered by forest



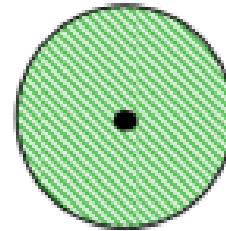
- Agriculture
- Agriculture (irrigated)
- Airport
- Bare rock
- Bare soils
- Forest
- Grassland
- Lake
- Riverbed
- Shrubs
- Urban centre
- Urban periphery



Forest := Con (Landuse="forest", "forest")

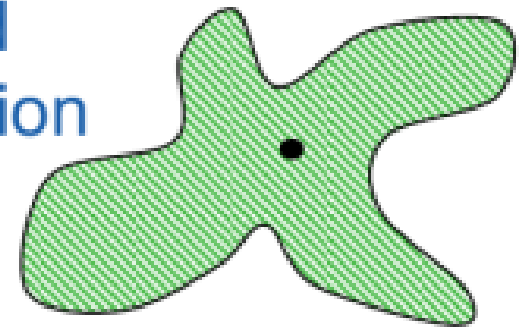
NEIGHBORHOOD FUNCTIONS

- **Proximity computation** makes use of the geometric distance function
- **Spread computation** assumes that the phenomenon spreads in all directions, but not necessarily equally easily in all direction
- In **Seek computation** the phenomenon will choose a least-resistance path

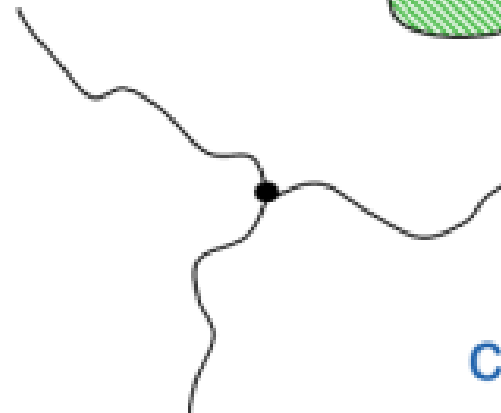


Proximity
computation

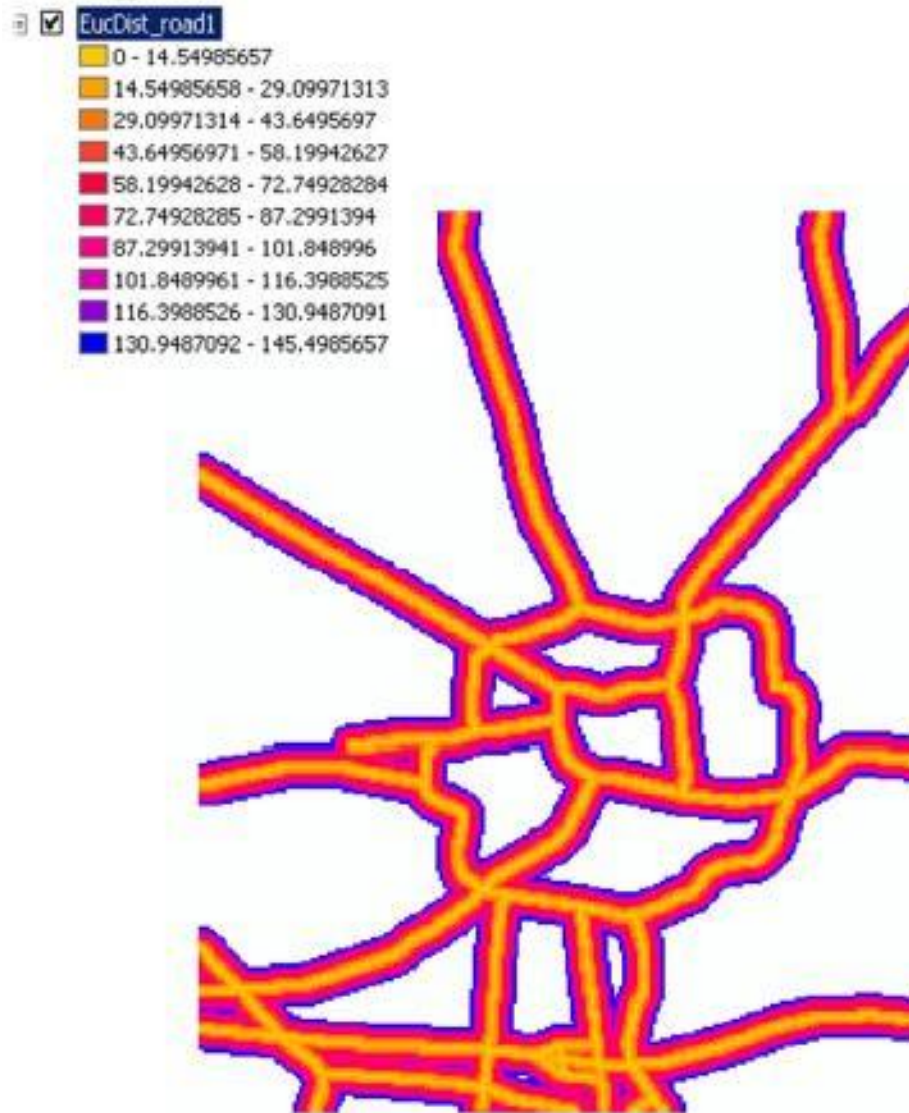
Spread
computation



Seek
computation

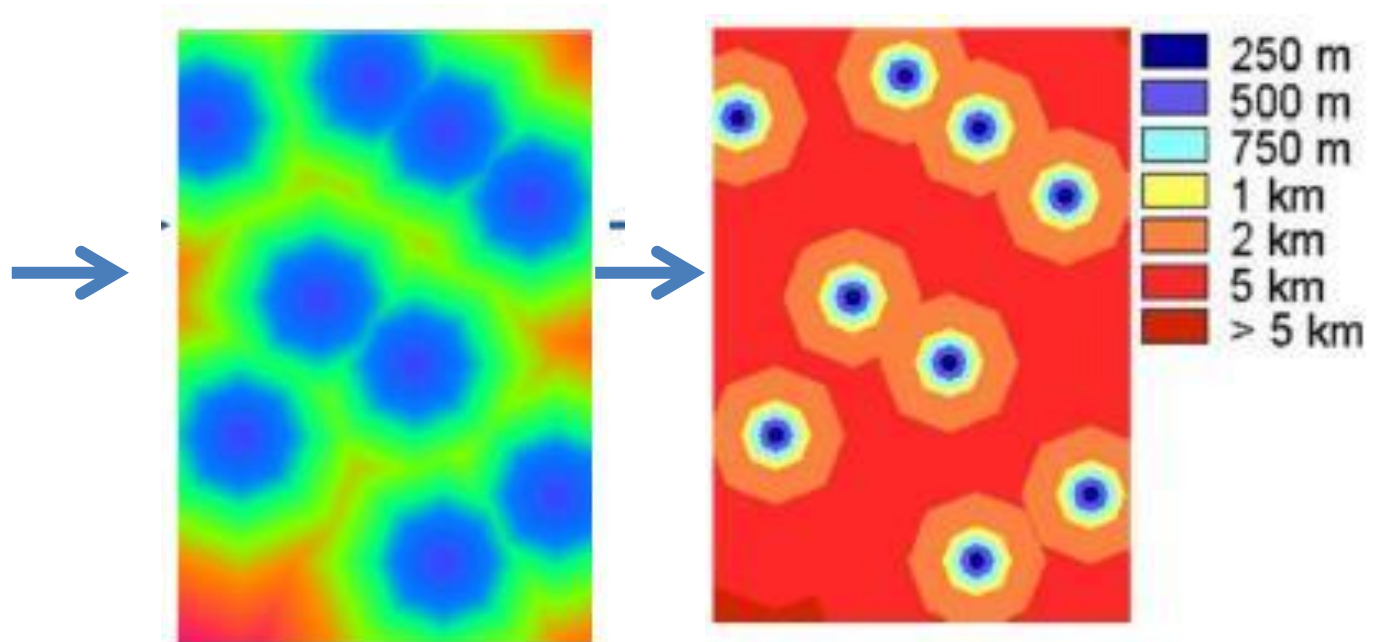


Proximity



Proximity

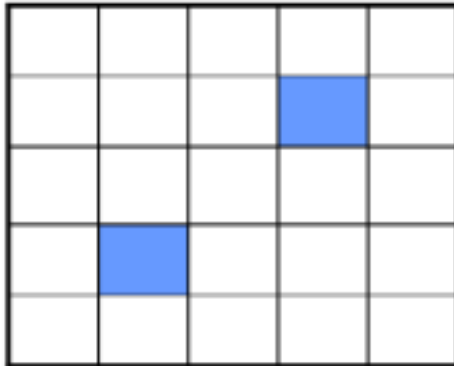
Location of wells



Spread computation

- The spread computation calculates a target location depends on **distance**, and the **direction**.
- Target locations contain a “**source material**” that spreads over time
 - Water and soil contaminants
 - Air (temperature, humidity)
 - People
- Typical examples where spread computation is useful:
 - Determination of inundated area due to dam construction
 - Determination of flooded area due to dike burst
 - Spreading of pollution
 - Commuters leaving a train station

Spread computation



Source locations

A high value,
indicates that it
is very difficult
to pass this cell

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

Local resistance raster

(high
resistance)

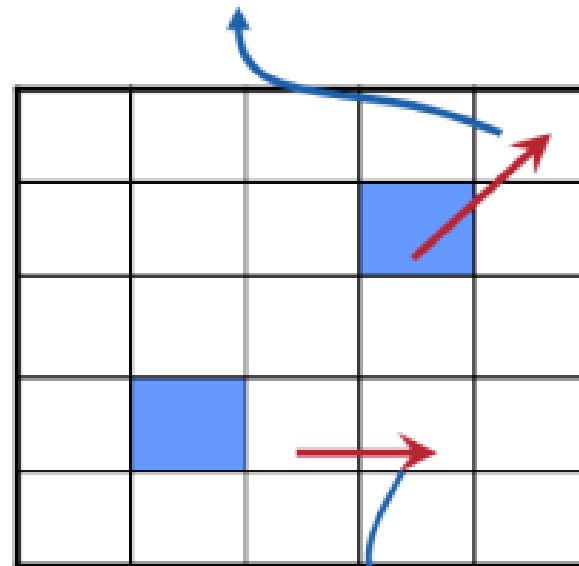
Spread computation involves

- one or more target locations, which are better called **source locations**
- a **local resistance raster**, which for each cell provides a value that indicates how difficult it is to pass by that cell.

Spread computation

- While computing total resistance, a GIS takes proper care of correct spread **path lengths**.
- The spread from a cell to its neighbor cell to the East is shorter than to its North-East neighbor

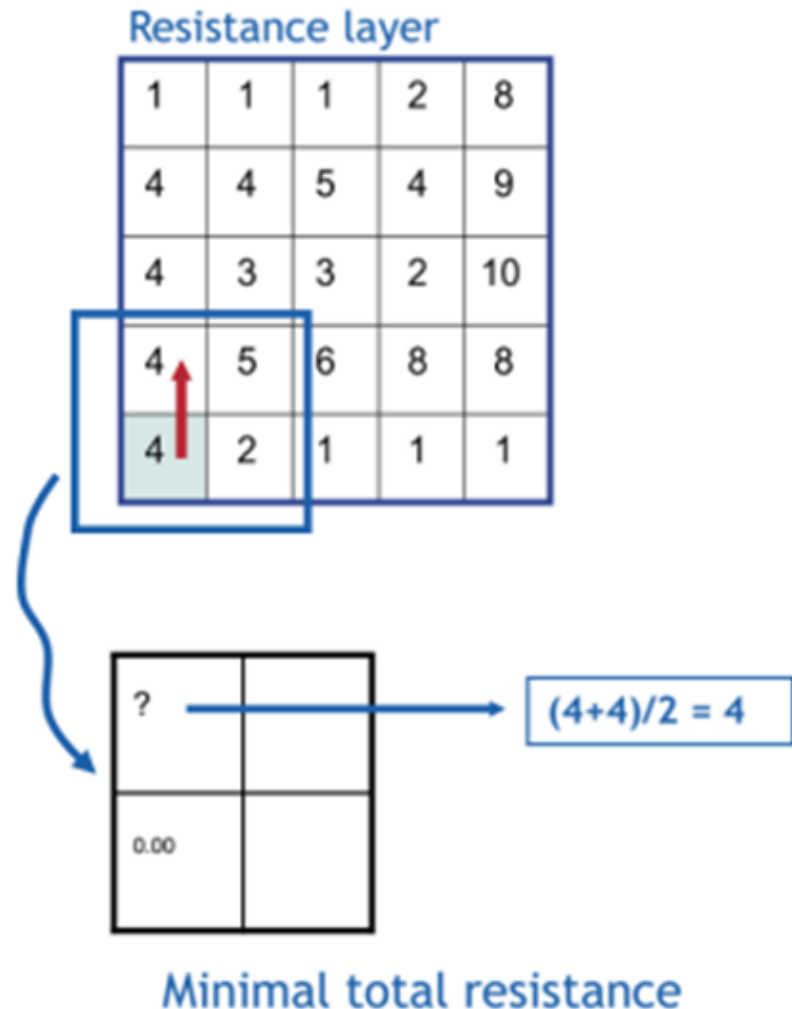
Path length to the North
East is cell size * $\sqrt{2}$



Path length to the East is
equal to the cell size

Spread computation

- A GIS computes the total resistance for spreading from c_{src} to c_n as
 $\frac{1}{2}(\text{val}(c_{src}) + \text{val}(c_n))$
- This is half of the resistance value of the source cell (c_{src}), because this cell is only traveled half, plus half of the resistance value of the cell North of it
- The value of the source cell is always 0.



Spread computation

- A GIS computes the total minimal resistance raster for a diagonal neighbor as

$$\frac{1}{2} (\text{val}(c_{\text{src}}) + \text{val}(c_{\text{ne}})) * \sqrt{2}$$

Resistance layer

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

4.00	?
0.00	

$$(4+5)/2 * \sqrt{2} = 6.36$$

Minimal total resistance

Spread computation

- Since spreading from a source material usually follows the easiest or shortest route, the value of minimal resistance is always determined.
- To determine the minimal resistance, a GIS calculates the resistance at all possible paths to reach a certain cell and then takes the minimal value.

Resistance layer

1	1	1	2	8
4	4	5	4	9
4	3	3	2	10
4	5	6	8	8
4	2	1	1	1

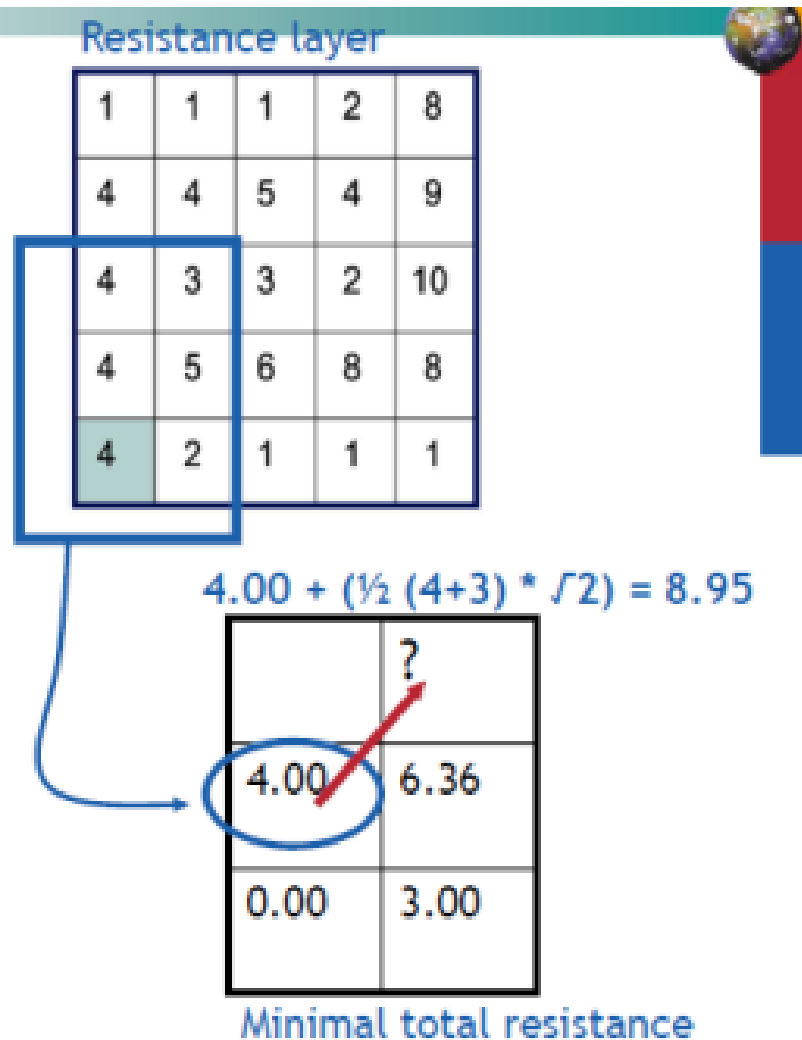
Total resistance via the black path $(4+5)/2 * \text{sqrt}(2) = 6.36$

Total resistance via the blue path $(4+4)/2 + (4+5)/2 = 8.5$

Total resistance via the red path $(4+2)/2 + (2+5)/2 = 6.5$

Lowest value

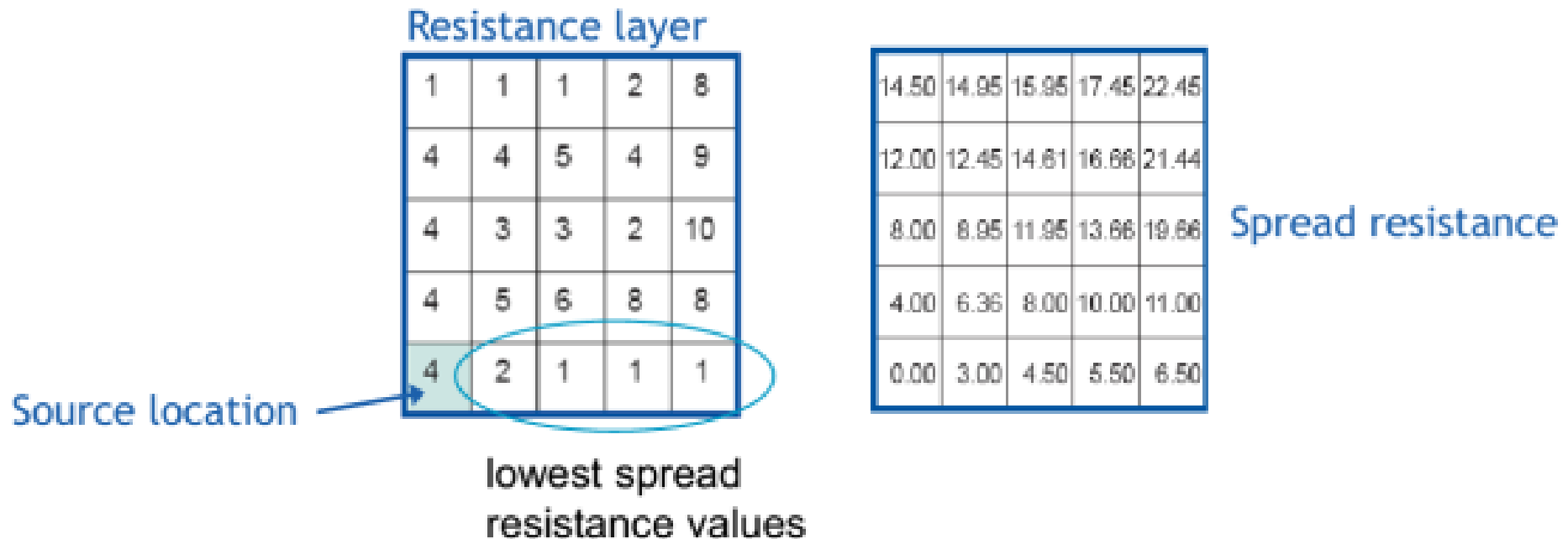
Spread computation



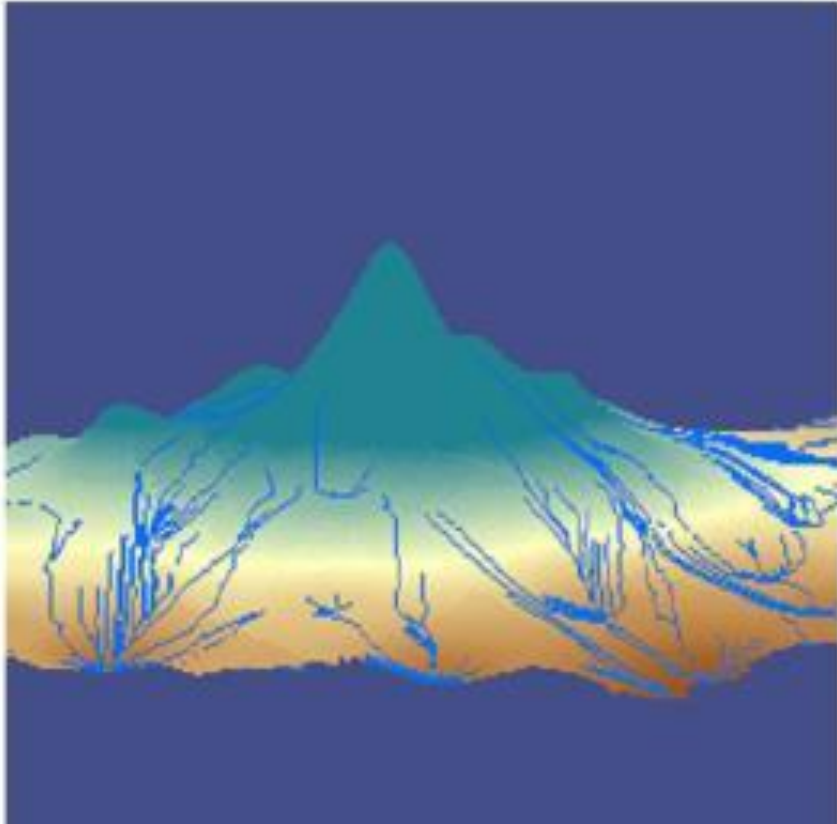
- The accumulated resistance along a path of cells is simply the sum of the resistances.
- We can use the value 4.00 and add the resistance for moving from this cell to its North East neighbor cell

Spread computation

- In this case, the source spreads more towards the lowest row of cells



Flow computation

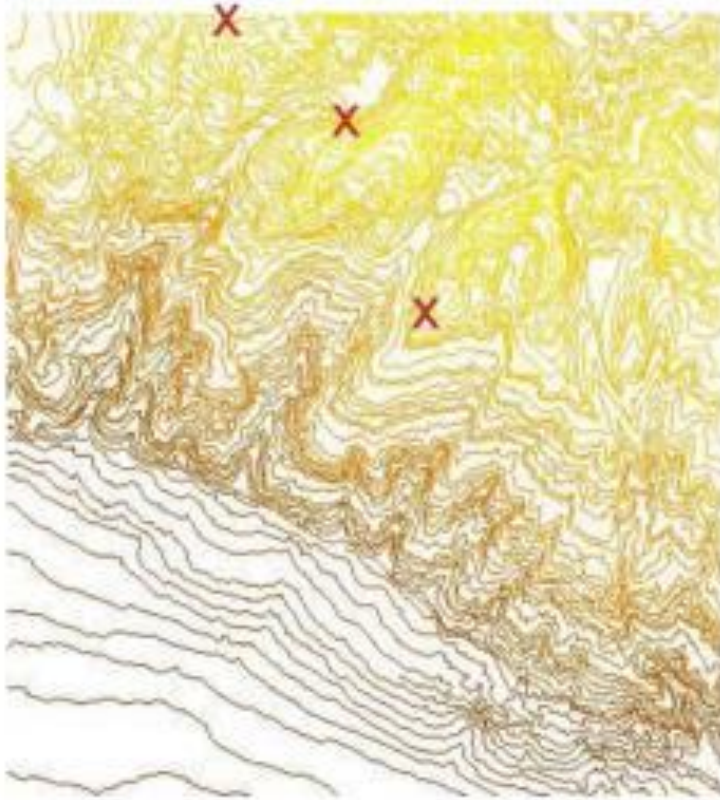


Flow computation

- Also called **seek** computation
- applies when a phenomenon does not spread in all directions, but chooses a **least-cost path**.
- Typical example: Determination of drainage patterns in a catchment

Flow computation

x: starting points



Automated flow path generation



Flow computation

Input: Elevation

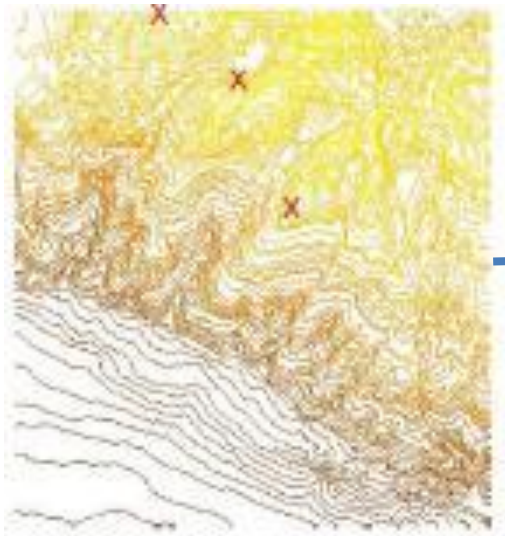
156	144	138	142	116	98
148	134	112	98	92	100
138	106	88	74	76	96
128	116	110	44	62	48
136	122	94	42	32	38
148	106	68	24	22	24

Output: Flow direction

↘	↘	↘	↓	↓	↗
↘	↘	↘	↓	↓	↗
→	→	↘	↓	↗	↓
↗	↗	→	↘	↓	↗
↘	↘	→	↓	↓	↓
→	→	→	→	↓	←

- Input is an **elevation raster**
- For each cell, the **steepest downward slope** to a neighbor cell is determined
- The direction of this downward slope is stored in the **flow direction raster**

Flow computation



156	144	138	142	116	98
148	134	112	98	92	100
138	106	88	74	76	96
128	116	110	44	62	48
136	122	94	42	32	38
148	106	68	24	22	24



156	144	138	142	116	98
148	134	112	98	92	100
138	106	88	74	76	96
128	116	110	44	62	48
136	122	94	42	32	38
148	106	68	24	22	24



↘	↘	↘	↓	↓	↙
↘	↘	↘	↓	↓	↙
→	→	↘	↓	↙	↓
↗	↗	→	↘	↓	↙
↘	↘	→	↓	↓	↓
→	→	→	→	↓	←

$$142 - 138 = 4 \quad 4$$

$$142 - 112 = 30 \quad 30 / \sqrt{2} = 21.2$$

$$142 - 98 = 44 \quad 44$$

$$142 - 92 = 50 \quad 50 / \sqrt{2} = 35.4$$

$$142 - 116 = 26 \quad 26$$

Flow computation

- From a flow direction map, a **flow accumulation map** can be derived.
- The accumulated flow for each cell is **the number of cells that flow into this particular cell**.
- Cells with a **value 0** have no other cells flowing into them, and represent **higher areas**.
- Cells with a **high accumulated flow** count represent **streams**.

Input: Flow direction

↘	↘	↘	↓	↓	↗
↘	↘	↘	↓	↓	↗
→	→	↘	↓	↗	↓
↗	↗	→	↘	↓	↗
↘	↘	→	↓	↓	↓
→	→	→	→	↓	←

Output: Flow accumulation

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

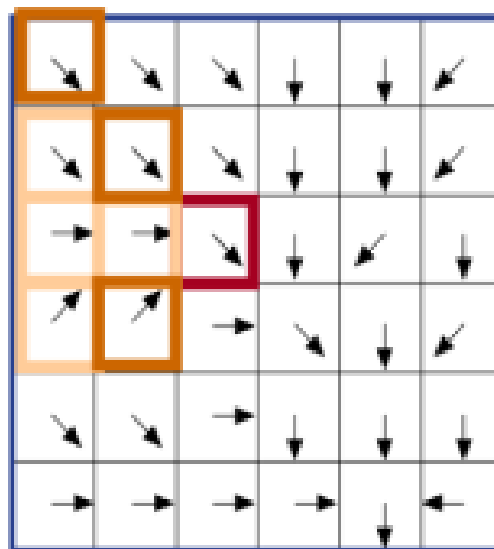
Flow computation

The number of cells that flow into this particular cell

156	144	138	142	116	98
148	134	112	98	92	100
138	106	88	74	76	96
128	116	110	44	62	48
136	122	94	42	32	38
148	106	68	24	22	24

(a)

Elevation



(b)

Flow direction

0	0	0	0	0	0
0	1	1	2	2	0
0	3	7	5	4	0
0	0	0	20	0	1
0	0	0	1	24	0
0	2	4	7	35	1

(c)

Accumulated flow count

4.4 Surface Modeling Operations

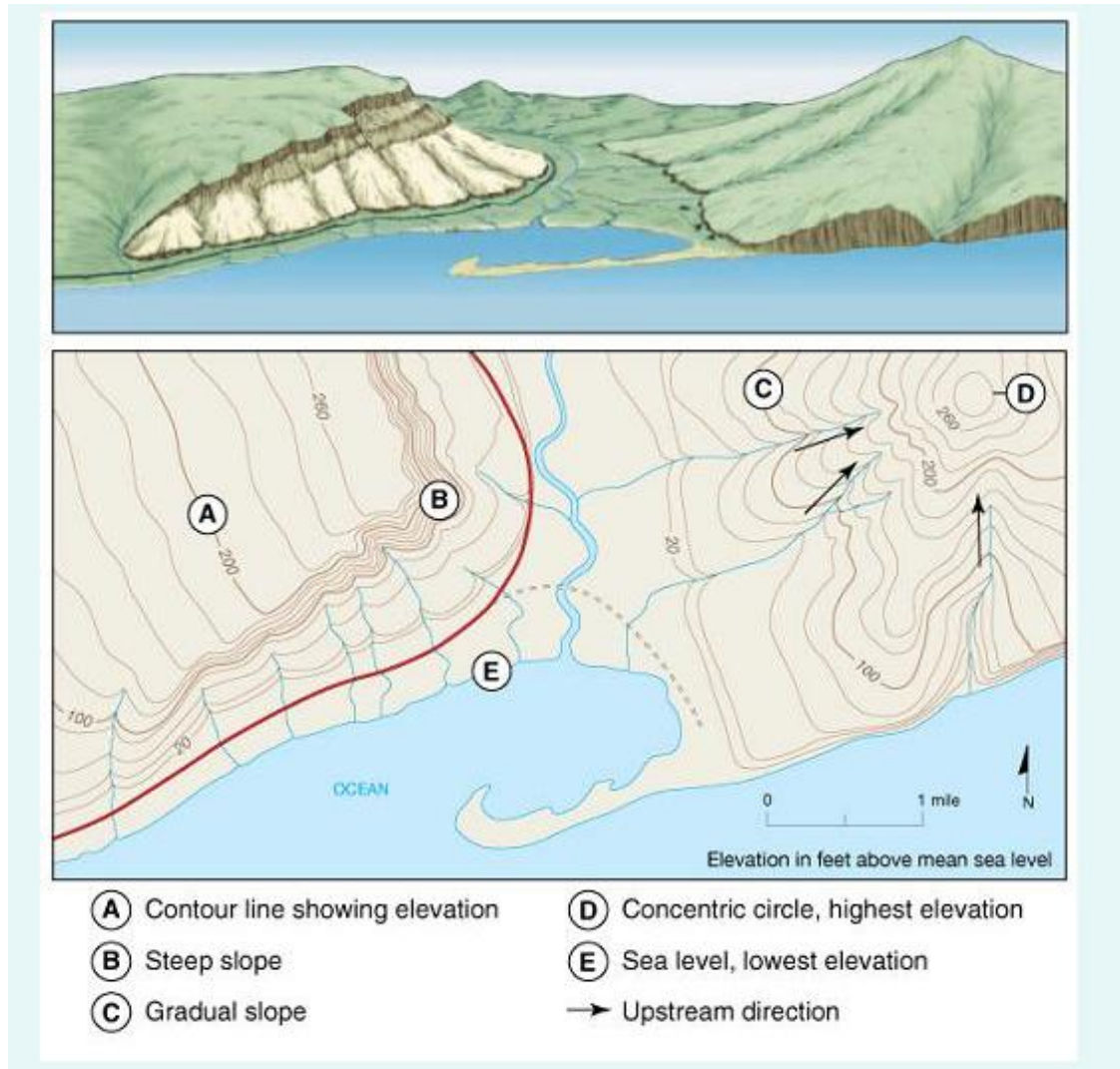
Surface modeling operations include:

- * Contour
- * Slope
- * Aspect
- * Hillshade
- * Viewshed
- * Visibility
- * Interpolation

Contour

- Contours represent points having equal heights/ elevations with respect to a particular datum such as Mean Sea Level (MSL)

- * Topographic maps, shape of a landscape represented with contour lines
- * A way to describe three-dimensional landscapes in two dimensions



Contour

- **Contour tool:** Creates a line feature class of contours (isolines) from a raster surface.
- A base contour is used; e.g. to create contours every 10 meters, starting at 5 meters
- 5 is the base contour, and 10 is the contour interval.
values to be contoured 5, 15, 25, 35, 45, ...

Slope

- Identifies the slope (gradient, or rate of maximum change in z-value) from each cell of a raster surface.

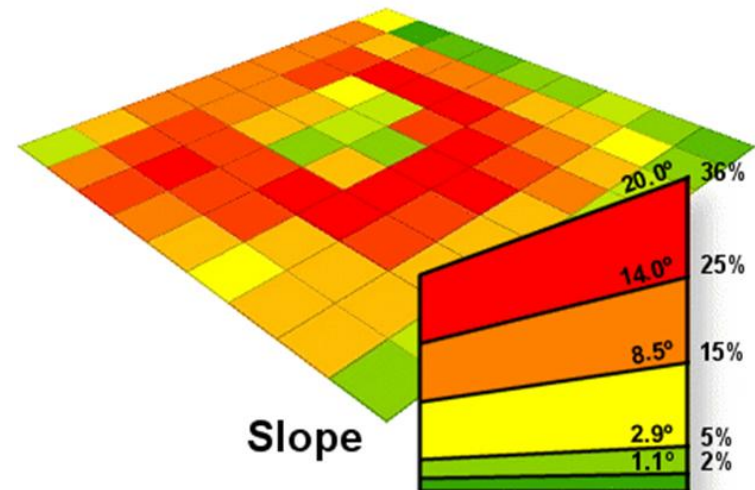
1	1	1	1	1	1
1	3	3	2	1	
1	1	3	2	2	2
1	2	2	2	2	2
1	1	1	2	2	2
1	1	1	1	1	2

InRas1

=

19.5	38.3	41.5	29.2	10.0	0.0
26.6	38.3	32.5	41.5	29.2	
21.6	43.8	21.6	26.6	14.0	10.0
14.0	29.2	29.2	14.0	0.0	0.0
10.0	21.6	29.2	29.2	21.6	10.0
0.0	0.0	10.0	21.6	29.2	14.0

OutRas



Aspect

- Derives aspect from a raster surface. The aspect identifies the **downslope direction** of the **maximum rate of change** in value from each cell to its neighbors.
- Aspect can be thought of as the slope direction. The values of the output raster will be the **compass direction** of the aspect.
- Aspect is expressed in positive degrees from 0 to 359.9, measured clockwise from north.

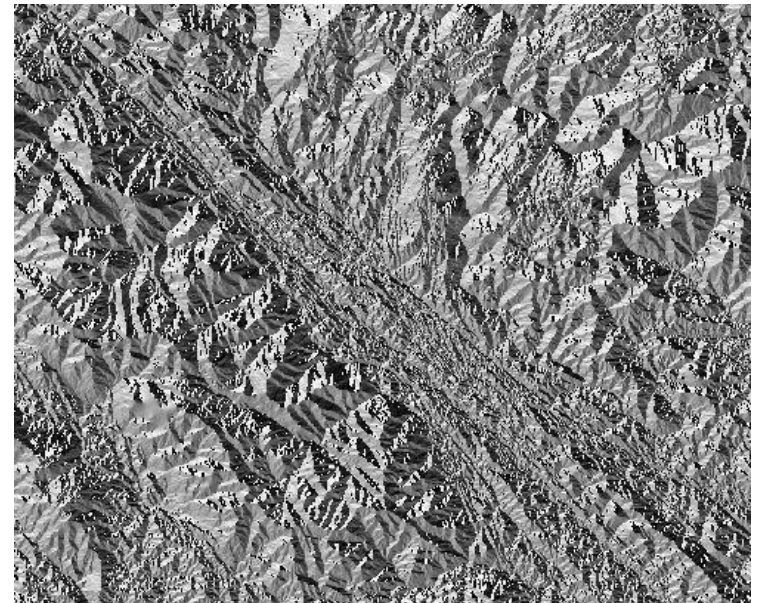
1	1	1	1	1	1
1	3	3	2	1	
1	1	3	2	2	2
1	2	2	2	2	2
1	1	1	2	2	2
1	1	1	1	1	2

InRas1

=

315.0	341.6	8.1	26.6	45.0	-1.0
270.0	266.4	11.3	45.0	26.6	
251.6	246.8	198.4	90.0	0.0	315.0
270.0	243.4	206.6	180.0	-1.0	-1.0
225.0	198.4	206.6	206.6	198.4	225.0
-1.0	-1.0	225.0	198.4	206.6	270.0

OutRas



Hillshade

- **Hillshade tool:** Creates a shaded relief from a surface raster by considering the illumination source angle and shadows.
- The hillshade raster has an integer value of 0 to 255

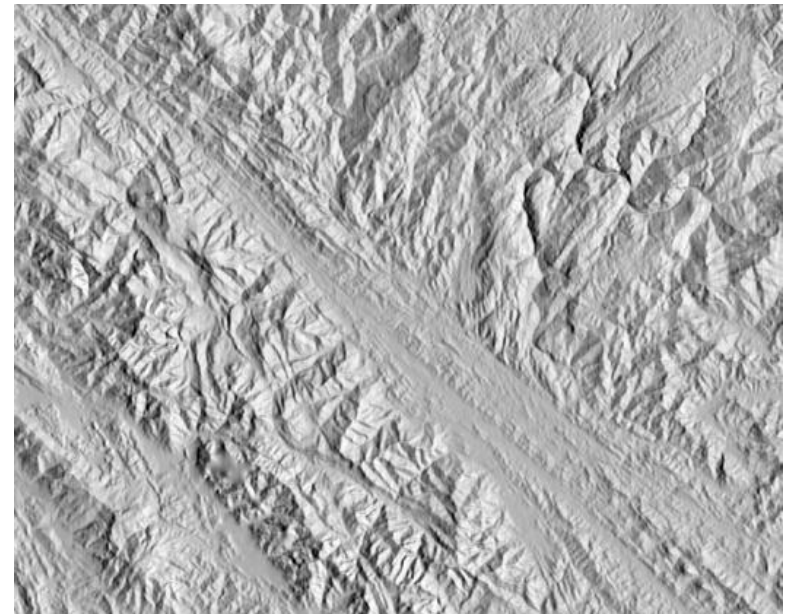
1	1	1	1	1	1
1	3	3	2	1	
1	1	3	2	2	2
1	2	2	2	2	2
1	1	1	2	2	2
1	1	1	1	1	2

InRas1

=

	0	121	187		
	0	118	218		
	36	89	142	136	
	116	89	89	115	

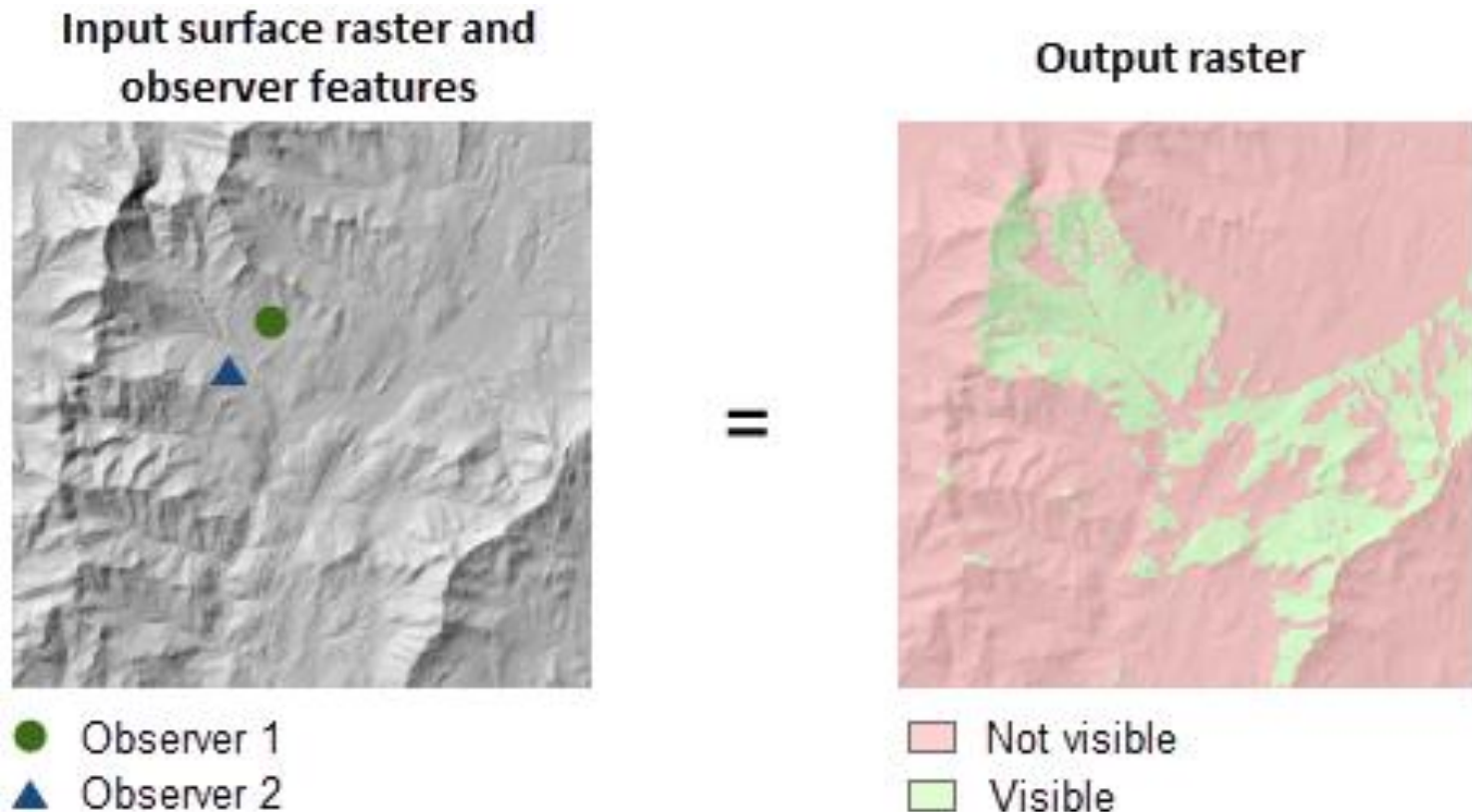
OutRas



$$\text{Hillshade} = 255.0 * ((\cos(\text{Zenith_rad}) * \cos(\text{Slope_rad})) + (\sin(\text{Zenith_rad}) * \sin(\text{Slope_rad}) * \cos(\text{Azimuth_rad} - \text{Aspect_rad})))$$

Viewshed

- Determines the raster surface locations visible to a set of observer features
- Useful in application like telecommunication

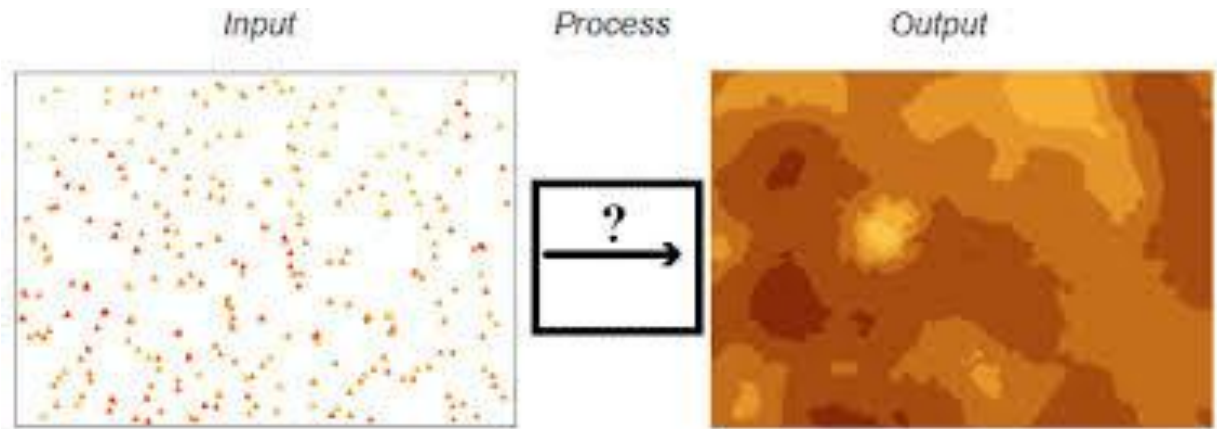


Spatial interpolation

- **Spatial interpolation** is used to take known values and interpolate them into a surface, deriving new estimated surface values
 - Elevation points interpolated to generate DEM
 - Contour lines interpolated to generate DEM
- Using interpolation methods, users can create surfaces from sampled locations without having to visit every location of a study area, saving time and effort.

Interpolation

- Interpolation is creating a surface based on a sample of values with the domain
- Different techniques for interpolation:
 - IDW (Inverse Distance Weighting)
 - Kriging
 - Natural Neighbor
 - Spline



IDW interpolation

- IDW: Interpolates **a raster surface from points** using an inverse distance weighted (IDW) technique
- Values for nearby points tend to be more similar
- IDW weights the value of each point by **its distance** to the cell being analyzed and averages the values
- It assumes that unknown value is influenced more by nearby points than far away points

Spline Interpolation

- **Spline:** Interpolates a raster surface from points using a two-dimensional minimum curvature spline technique.
- It fits a curve through the sample data and assigns values to other locations based on their location on the curve
 - Regularized Method: results in a smoother surface that smoother areas of abruptly changing values
 - Tension Method: results in a rougher surface that more closely adheres to abrupt changes in sample points

Kriging Interpolation

- Kriging Method: Interpolates a raster surface from points using kriging.
- Like IDW interpolation, Kriging forms weights from surrounding measured values to predict values at unmeasured locations.
 - IDW uses a simple algorithm **based on distance**
 - Kriging weights come from a semi-variogram that is developed by looking at the **spatial structure** of the data
 - Predictions are made for locations in the study area based on the semi-variogram and the spatial arrangement of measured values that are nearby
 - Semi-variograms measure the strength of statistical correlation as a function of distance; they quantify spatial autocorrelation
 - Kriging associates probability with each prediction

Practice 2. Spatial analysis (Vector)

Data: North communes_Vietnam.shp

1. Measurement:
Distance, Area, Angles
2. Query
Select by geometry: Edit/ Select
Select features by attributes: Attribute table/ Select features using expression

- to select all communes in Hanoi with the area < 01 km2 and population > 15000
3. Calculating the population density
- open Attribute table: Field calculation
 input expression: POPULATION/AREA
- Classification using the field "Population density"
 Properties/ Symbolology
 Classify by Equal count (quantile) and Equal interval, then compare the results

Practice with spatial analysis (Vector)

5. Density interpolation

Purpose: Interpolate the PM2.5 in the north of Vietnam using PM2.5 measurement stations.

Inputs: PM2.5.csv, Map north of Vietnam

- Open files*
- Processing tools/Interpolation/IDW interpolation*
- Raster/ Extraction/ Contour*
- Processing tools/ Raster analysis/Zonal statistics (Statistic data to define the max and mean pollution for every commune)*

6. Land cover change analysis

Purpose: Define the changes of land cover during 2015-2017 and calculate the change area for each cover types.

Digitalizing land cover map 2015 with 8 objects (same to Landcover 2017)

Overlay land cover map in 2015 and in 2017

Analysis of changes

Editing change map

Steps: (Option 1)

- Create the change map: Vector/ Geoprocessing tools/ Intersection*
- Create change field: In Attribute table: Expression: "Type2015" + '_' + "Type2017"*
- Re-calculate the change area*
- Export attributable to excel (In Attribute table, Copy selected rows to clipboard)*
- Create the change table in Excel*

Practice with spatial analysis (Vector)

Steps: (Option 2)

- *Convert land cover 2017 to raster: Raster/ Conversion/ Rasterize/ Vector to Raster (must change the character to number type)*
- *Calculate the change area (intersection between raster and vector): Processing toolbox/ Raster analysis/Zone histogram*
- *Open the vector output file to see the changed pixels.*
- *Calculating area from changed pixels by multiply the number of change pixels with the area of a pixel*

7. Define the location to build a market

Criteria: Distance between houses and the market is smaller than $< 130\text{m}$, and the market is far from electrical wire $> 30\text{m}$?

- *Draw houses, electric lines*
- *Create buffer for Houses (radius 130 m), Electric lines (30 m)*
- *Save Buffer of each house in individual layer.*
- *Intersect buffers of individual layers of 3 houses.*
- *“Difference” between Buffer of houses and Buffer of electric lines*

Example of change table

Change area (m2)		Land cover 2017								Sum 2015
		Residence	Cemetery	Road	Forest	Annual crop	Perenial trees	Rice	Fish pond	
Land cover 2015	Residence	28100.20		4216.38	1858.36	139.35			362.80	34677.09
	Cemetery		24889.21				1472.84		1074.92	27436.96
	Road	1340.67		5446.01						6786.69
	Forest	7640.95			20438.65					28079.59
	Annual crop	4151.54		210.51						4362.04
	Perenial trees	4202.45	94.36	307.75					2118.77	6723.34
	Rice	4581.90		788.38			2578.69	8954.35	2152.70	19056.01
	Fish pond	14581.67	2713.92	4198.28		9773.81	1083.53	90.64	38737.28	71179.12
Sum 2017		64599.37	27697.50	15167.31	22297.01	9913.16	5135.05	9044.98	44446.46	

Practices 3.

Data: HoaBinh_HaNoi_DEM.tif

1. Classification: equal interval and equal frequency

(Raster analysis/ Reclassify by table)

- Equal interval classification
- Quantile classification

2. Surface modeling operations:

(Raster terrain analysis)

- Slope
- Aspect: Define the slope direction according to North, West, South, East
- Hillshade
- Contour *(Raster/ Extraction/ Contour)*

Problem: Determine the appropriate areas for planting café to get the potential productivities.

(Café: slope 10° - 30° , slope direction NE-E-SE, elevation 1000-1500 m)

3. Flow directions (Extracting a stream network from DEM)

Input: DEMs

Outputs: Watersheds, Channel networks

Preprocessing: Processing Tab > Toolbox > SAGA>Terrain analysis-preprocessing

1/ Fill Sinks (Wing & Liu): *“To correct the DEM”*

Processing Tab > Toolbox > SAGA > Terrain Analysis-Channels

2/ Channel Network and Drainage Basins

Catchment Area: *“The water flows over topographic terrain and consists of water runoff into surrounding rivers, streams and lakes”*

https://gracilis.carleton.ca/CUOSGwiki/index.php/Exploring_the_Hydrological_Tools_in_QGIS

Practices 4. Flooding risk model

- Purposes: Determine the area of flood risks.
- Input data: Slope, River density, Elevation stratification, Land cover
- Output: Potential flooding
- Methods: Multi criteria analysis (MCA)

(MCE bases on the assumption: the potential of flooding is effected by the slope, river density, elevation, land cover, infiltration rate,)

$$F = 0.45 * \text{Slope} + 0.4 * \text{Land cover} + 0.15 * \text{Elevation}$$

- Steps

1/ Classifying Slope into: (*Raster analysis/ Reclassify by table*)

(0-5°) = High risk ranked as 10

(5°-20°) = Medium risk ranked as 5

(20°-70°) = Low risk ranked as 2

Step 3/ Elevation (*Raster analysis/ Reclassify by table*)

0 m – 100 m = High risk ranked as 10

100 m – 500 m = Medium risk ranked as 5

500 m - 3000 m = Low risk ranked as 2

Step 4/ Reclassifying land cover map (*Raster analysis/ Reclassify by table*)

Download land cover map: <https://lcviewer.vito.be/download>

<https://zenodo.org/records/4723921#.YWWKz9IBzPY>

Clip land cover map following the Hanoi_HoaBinh_DEM: (Raster/Extraction/ Clip raster by extent)

Reclassify:

(Bare land, residence) = High risk ranked as 10

(Shrubs, agriculture land) = Medium risk ranked as 5

(Sparse forest) = Low risk ranked as 2

(Closed/ Dense forest) = Very low risk ranked as 1

Step 5/ Calculate the potential of flood (*Raster calculator*)

$$F = 0.35 * \text{Slope} + 0.3 * \text{Land cover} + 0.2 * \text{River density} + 0.15 * \text{Elevation}$$

Step 6/ Rank the risk of flood (*Raster analysis/ Reclassify by table*)

High risk = 7 ; 10 as 4

Medium risk = 5 ; 6 as 3

Low risk = 3 ; 4 as 2

No risk = 0 ; 2 as 1

Global land cover map

20	A12A4A20B3(B9)	Shrubs	These are woody perennial plants with persistent and woody stems and without any defined main stem being less than 5 m tall. The shrub foliage can be either evergreen or deciduous.	255, 187, 34
30	A12A2(A6)A20B4	Herbaceous vegetation	Plants without persistent stem or shoots above ground and lacking definite firm structure. Tree and shrub cover is less than 10 %.	255, 255, 76
60	B16A1(A2)	Bare / sparse vegetation	Lands with exposed soil, sand, or rocks and never has more than 10 % vegetated cover during any time of the year	180, 180, 180
40	A11A3	Cultivated and managed vegetation/agriculture (cropland)	Lands covered with temporary crops followed by harvest and a bare soil period (e.g., single and multiple cropping systems). Note that perennial woody crops will be classified as the appropriate forest or shrub land cover type.	240, 150, 255
50	B15A1	Urban / built up	Land covered by buildings and other man-made structures	250, 0, 0

Practices 5. Network analysis

- **Purposes:** Calculating the shortest or fastest path between two points and determining Service area.
- **Tools:** QGIS; Processing / Network Analysis
- **Data:** Roads_north_Vietnam.shp

1/Calculate the shortest path (point to point)

Processing / Network Analysis/Shortest path (point to point)

- *Select on the screen the: Road file, Starting point, End point*
- *See the attribute table for the length*

(alternatively selecting “Fastest” in the “Path type to calculate”. In “Advance parameters” select the “Speed field”)

2/ Estimate the “Service area”

Processing / Network Analysis/Service area (from point)

- *Select on the screen the: Road file, Starting point, alternatively selecting “Fastest” or “Shortest”, Travel cost*
- *See the results*

3/ Estimate the Service area (from layer)

- Starting points are defined in a point layer